Do Fusion Operators Matter, and Does the Answer Survive Preprocessing? A Parameter-Matched Multiverse Comparison of Multimodal CNN Fusion for EEG Seizure Detection

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DRAFT v2, NOT FOR SUBMISSION. Restructured around the three research questions the advisor set on 10 October 2026. This file holds the Introduction (Section 1), the Methods (Section 2, in the advisor's subsections 2.1 to 2.7) and the Results of the five-seed runs (Section 3) and the Discussion (Section 4). The literature review is provisional: the systematic search has run, but title and abstract screening was done by an AI assistant, and the human screening and the full texts follow after 17 October; claims marked [abstract] rest on a source's abstract only and are to be checked against its full text. `paper/manuscript.md` is the previous draft (Bonn and CHB-MIT, single pipeline) and stays unchanged until this one replaces it; the Bonn benchmark moves to Supplementary Note 1 (advisor, 16 October 2026). Every number below is traceable to a file in the repository (paths given).

Abstract

Studies of convolutional networks for EEG seizure detection that fuse the raw waveform with its spectrogram often report that one fusion operator outperforms the others, on one dataset, one preprocessing pipeline and one training run. We tested whether such conclusions hold when these choices change. Five fusion operators (early, late, gated and attention fusion, matched in parameter count, and score fusion, an ensemble of the two single-branch networks) and single-branch and classical controls were compared under nine preprocessing pipelines, including three artefact removal algorithms (Extended Infomax, GEDAI, AMICA), and five seed sets, with leave-one-subject-out evaluation on CHB-MIT (23 subjects) and Siena (14 subjects) and by training on CHB-MIT and testing on Siena. Statistical significance (resampling-corrected tests), practical equivalence (margins fixed in advance) and ranking stability were assessed separately. The operators differed by a median of 0.021 (CHB-MIT) and 0.053 (Siena) macro F1 within a pipeline, and no difference was significant. Variability associated with the held-out subject made up 74 percent of the variance on CHB-MIT; the estimated shares of preprocessing and the operator were zero, that of the seed 1 percent. Rankings changed between seeds (median Kendall's W 0.22 on CHB-MIT), were more stable on Siena, where score fusion ranked first in 82 percent of runs without a significant advantage, and did not transfer between datasets. Early fusion had the lowest AUPRC in all 18 pipeline-dataset combinations. The controls stood to the operators as the operators stood to each other: a spectrogram network of the same size and a logistic regression on seven hand-crafted features reached similar or higher mean macro F1 at the 0.5 threshold, the latter above the best operator in 43 of 45 Siena runs and the highest there at every threshold tested, but no difference was significant. With fixed seeds and package versions, GEDAI's threshold-based cleaning was deterministic on each machine but not across machines: it cleaned 27 percent of recordings differently on a second machine, enough to change seizure-level sensitivity by up to 0.054 and the order of the operators. Claims that one fusion operator is superior, made from a single dataset, pipeline and run, are not supported.

1. Introduction

Electroencephalography (EEG) records the brain's electrical activity, and during an epileptic seizure that activity changes characteristically: amplitude rises, the signal becomes rhythmic and its power concentrates in particular frequency bands. Automatic seizure detection is therefore a signal classification problem, and convolutional neural networks (CNNs) are the dominant approach to it. A recurring design represents each EEG segment twice, as the raw waveform for a 1D CNN and as its time-frequency image for a 2D CNN, and combines the two branches with a fusion operator: early fusion of inputs or early features, late fusion of the final feature vectors, a learned gate or attention weighting, or a combination of the two branches' predicted scores. Studies of such models typically report that one operator outperforms the others. Whether such a statement holds once the analytic decisions around the comparison change is the question of this paper.

1.1 Preprocessing changes what deep models achieve

Deep networks are often expected to learn from minimally processed EEG, yet the choice of preprocessing changes their results. In the first systematic study of EEG preprocessing for deep learning, 4,800 models trained on six tasks with four architectures differed significantly between preprocessing pipelines; models trained on raw data ranked last on average, and minimal pipelines without artefact handling tended to serve them best (Del Pup et al. 2025) [abstract]. Artefact removal by independent component analysis, with Infomax or AMICA and three component rejection strategies, did not consistently improve deep decoding on three brain-computer interface datasets (Kang et al. 2024) [abstract], and even the scope of normalisation changed which strategy worked best (Truong and Delorme 2025) [abstract]. Outside deep learning the same sensitivity is documented by multiverse analyses, which run and report every defensible combination of analysis choices instead of one (Steegen et al. 2016; Clayson 2024): across 43 preprocessing pipelines no single pipeline was best on every quality criterion (Huang et al. 2025) [abstract]; fourteen pipelines drawn from the N400 literature changed statistical outcomes, effect sizes and power (Šoškić et al. 2024) [abstract]; and only 13 of 270 analyses of frontal alpha asymmetry in depression were significant (Kołodziej et al. 2021) [abstract]. For event-related potentials, multiverse analyses have been used to choose pipelines by their effect on data quality and reliability (Clayson et al. 2021) [abstract]. A comparison of fusion operators made under one preprocessing pipeline therefore cannot tell whether its ranking belongs to the operators or to the pipeline.

1.2 Fusion operators are compared, but not under controlled conditions

Several recent studies compare fusion operators for EEG seizure detection or prediction directly (Table 1): five strategies within and across patients (An et al. 2026), four operators under leave-one-subject-out evaluation, where they lie within one standard deviation of each other (Einizade et al. 2023), three operators with paired tests over six seeds (Basheer and Mishra 2026), and feature-level against decision-level fusion for seizure prediction (Daşdemir and Örnek 2025). Among the studies we have read in full, none matches the parameter counts of the compared operators, tests the differences between them with a correction for repeated resampling, and tests them for equivalence; a larger fusion model that beats a smaller one may owe its advantage to capacity rather than to the way it fuses. Across more than 120 studies of hybrid EEG interfaces, only two explored more than one fusion strategy and one asked which was best (Lee et al. 2025). Controlled comparisons in other domains point the same way: in ECG classification 1D and 2D representations were compared with early, late and joint fusion (Narotamo et al. 2024), a controlled benchmark on tabular neuroimaging data found the fusion strategy to matter more

than the architecture (Camastra et al. 2026), and added multimodal complexity often did not pay off (Rheude et al. 2025).

Table 1: Closest related work (from the systematic search of 3 October 2026; studies in bold are the closest to this one; the full texts of the remaining candidates are still being read).

Work	Contribution	What it leaves open
An et al. (2026), fastSeizureNet	Compares five fusion strategies for EEG seizure detection, within and across patients, with several backbones	Fuses hand-crafted with deep features rather than a waveform with its spectrogram; strategies not parameter matched and not tested against each other (a Wilcoxon test with Holm correction is used only against prior methods)
Einizade et al. (2023), fAttNet	Compares four fusion operators over raw EEG, wavelet packet and hand-engineered views, leave-one-subject-out on TUH	No parameter matching, no statistical test; the operators lie within one standard deviation of each other
Basheer and Mishra (2026)	Compares concatenation, element-wise addition and KAN gating, paired t tests over six seeds	Both streams come from the same raw EEG; uncorrected tests against concatenation only; no parameter matching
Daşdemir and Örnek (2025)	Protocol-controlled comparison of two fusion strategies for EEG	Seizure prediction, not detection; two operators; a 0.20 point accuracy difference without a stated test or parameter matching
Wang et al. (2026), *Neurocomputing*	Cross-attention fusion of time-domain and S-transform views, CHB-MIT and Siena	One ablation against concatenation, patient-specific evaluation, no statistical test
Chatzichristos et al. (2020)	Attention-gated multi-view fusion	One operator, no alternatives
Das et al. (2024); Huang et al. (2024)	1D+2D fusion; multi-head attention fusion	Fixed design, no operator ablation; gains without matched parameter counts
Narotamo et al. (2024)	1D and 2D representations with early, late and joint fusion	ECG; no parameter matching or corrected and equivalence statistics
Camastra et al. (2026), *Brain Sciences*	Controlled fusion benchmark: strategy matters more than architecture	Tabular neuroimaging features; no paired or equivalence testing
Ali et al. (2024)	CHB-MIT results depend heavily on evaluation choices	Not about fusion operators
Dan et al. (2024), SzCORE	Standard datasets, event-based scoring and metrics for seizure detection	An evaluation framework
Kontras et al. (2026), NeuroAtlas	Bonn is saturated (eleven models at AUROC >= 0.99) and has no subject identifiers	Not a fusion comparison

1.3 One seed or one pipeline is a weak basis for a ranking

A comparison built on one training run can be decided by factors unrelated to the methods compared. Identical training runs of deep networks produce different models: even after weak models were excluded, accuracy differed by up to 10.8 percent between identical runs, implementation-level factors alone caused differences of up to 2.9 percent, and only about one fifth of the surveyed papers used repeated runs to quantify this (Pham et al. 2020) [abstract]. Every source of nondeterminism in network optimisation, down to a one-bit change in the initial parameters, produces similar run-to-run variability, which makes small improvements hard to separate from noise (Summers and Dinneen 2021) [abstract], and seeds that perform far better or worse than average are easy to find (Picard 2021) [abstract]. The computing environment adds its own variance: CPU multithreading makes training nondeterministic (Xiao et al. 2021)

[abstract], floating-point non-associativity in parallel computation causes run-to-run differences (Shanmugavelu et al. 2024) [abstract], and reproducible training requires control of both software randomness and hardware nondeterminism (Chen et al. 2022) [abstract]. In speech recognition, training variance was large enough to call for a change in how results are reported (van den Berg et al. 2017) [abstract], and seed-induced variability can be tested for directly (Banerjee et al. 2025) [abstract]. The statistics add a further trap: under repeated cross-validation the training sets overlap, so standard paired tests are overconfident unless their variance is corrected (Nadeau and Bengio 2003; Bouckaert and Frank 2004), and whether a comparison comes out significant can depend on the number of folds and repeats (Jafrasteh et al. 2025).

1.4 Results on one dataset rarely carry over to another

High accuracies reported for automatic seizure detection often do not hold on new patients or recording sites. When 28 algorithms from the SzCORE challenge were evaluated on 4,360 hours of held-out continuous EEG from 65 subjects, the best F1 score was 32 percent, far below self-reported results, and the algorithms with the highest aggregate score were not the most consistent across subjects (Dan et al. 2025, preprint) [abstract]; the same group proposed standard datasets, cross-validation strategies and event-based metrics to make such comparisons possible (Dan et al. 2024). Real-time detectors had been tested in distinct settings until they were compared under one (Lee et al. 2022, preprint) [abstract]; a prospective test on about 14,590 hours of adult EEG from another continent reported 76.68 percent with about 56 false alarms per 24 hours (Yang et al. 2021, preprint) [abstract]; and a systematic review names generalisability across training data and acquisition hardware among the main obstacles to clinical translation (Moutonnet et al. 2024, preprint) [abstract]. Within one public corpus, CHB-MIT results depend heavily on evaluation choices (Ali et al. 2024), and the Bonn corpus, long a standard benchmark, is saturated and carries no subject identifiers, so it cannot test generalisation across patients (Kontras et al. 2026).

1.5 This study

Our contribution is not another fusion operator, nor only a comparison of existing ones: it is a test of how far conclusions drawn from such a comparison hold when the analytic decisions around it change. We hold the backbone, the training protocol and the parameter budget fixed (the four jointly trained operators within 0.4 percent; score fusion, an ensemble of the two branches, is compared alongside them but is not parameter-matched), vary only the fusion operator, and repeat the whole comparison under nine preprocessing pipelines, including three artefact removal algorithms, and five seed sets, on two independent corpora and from one to the other. We ask three questions.

- **RQ1, performance robustness.** Do fusion operators maintain their predictive performance across different EEG preprocessing pipelines?
- **RQ2, ranking robustness.** Are the rankings of fusion operators stable across preprocessing choices, artefact removal algorithms and stochastic training?
- **RQ3, cross-dataset generalisation.** Do fusion performance patterns and rankings generalise across independent EEG seizure datasets?

Throughout, three properties are kept apart, because each can hold without the others: **statistical significance** (is there evidence of a difference between two operators?), **practical equivalence** (does their difference lie within a margin fixed in advance?) and **ranking stability** (is their order preserved when preprocessing or seeds change?). In particular, the absence of a significant difference is not taken as evidence of equivalence.

The study contributes:

1. a comparison of five fusion operators, four of them jointly trained and matched in parameter count and the fifth a decision-level ensemble outside that matching, with single-branch and classical controls, crossed with nine preprocessing pipelines and five seed sets on CHB-MIT and Siena, and a transfer experiment from one to the other;
2. an analysis that separates the effects of preprocessing, seed and their interaction with the operator, and reports each operator's full rank distribution rather than a single ranking;
3. a test of the reproducibility of three artefact removal algorithms across machines and execution contexts, and of how much their irreproducibility changes the comparison itself;
4. corrections to the source data and its documentation: a labelling error in CHB-MIT's annotations, two of its cases that are one person and must be held out together, a mains notch that has to follow each dataset's line frequency, and six problems in Siena's seizure lists, each resolved and documented;
5. the code, the per-fold results of every run, and every analysis, public and regenerated by single commands.

2. Methods

Figure 1 summarises the design: the datasets, the nine preprocessing pipelines, the two representations, the fusion operators and controls, and the evaluation; its panel b shows where each step is applied and how leakage between training and test data is prevented.

2.1 Datasets and EEG harmonisation

CHB-MIT (Shoeb 2009; Goldberger et al. 2000) holds 686 scalp recordings, 982.9 h, from 24 paediatric cases at 256 Hz in a bipolar montage. Two cases, chb01 and chb21, are documented as the same person and are treated as one subject, which gives 23 subjects. We use the standard 18-channel bipolar (double banana) montage and keep the 683 recordings that contain all 18 channels, with 185 of the 198 annotated seizures in the windowed corpus; the three dropped recordings belong to part of subject chb12's data recorded in a different montage. Seizure annotations were parsed by order rather than by their printed number, which corrects a labelling error in `chb09_08.edf` (`src/chbmit.py: parse_summary`).

Siena Scalp EEG (Detti et al. 2020) holds 41 recordings, 141.0 h by the EDF headers, from 14 adult patients, with 47 annotated seizures, recorded at 512 Hz in a referential 10-20 montage (older electrode names T3 to T6 are mapped to T7, T8, P7, P8). Six problems in the published seizure lists (a seizure ending after its recording, two conflicting onset or end times, two start times that disagree with the EDF headers, an onset given only clinically) were resolved before any analysis, each as either a correction verified against the recording or a stated decision rule; Supplementary Table S4 lists every decision, how it was checked, its remaining uncertainty, and how much the alternative would change the windows (`docs/SIENA.md`).

Harmonisation. Siena is converted to the CHB-MIT form before anything else: the 18 bipolar channels are formed as electrode differences of the referential signals, and the continuous recording is resampled from 512 to 256 Hz with a polyphase filter (`scipy.signal.resample_poly`, factor 1/2; `src/siena.py: read_bipolar`). Both corpora then pass through identical code. Seizure times are converted to seconds from each recording's first sample using the EDF header start time (modulo 24 h for recordings past midnight).

Windows and labels. Recordings are cut into non-overlapping 10 s windows. A window is ictal if it overlaps an annotated seizure at all. Non-ictal windows are subsampled per subject to four

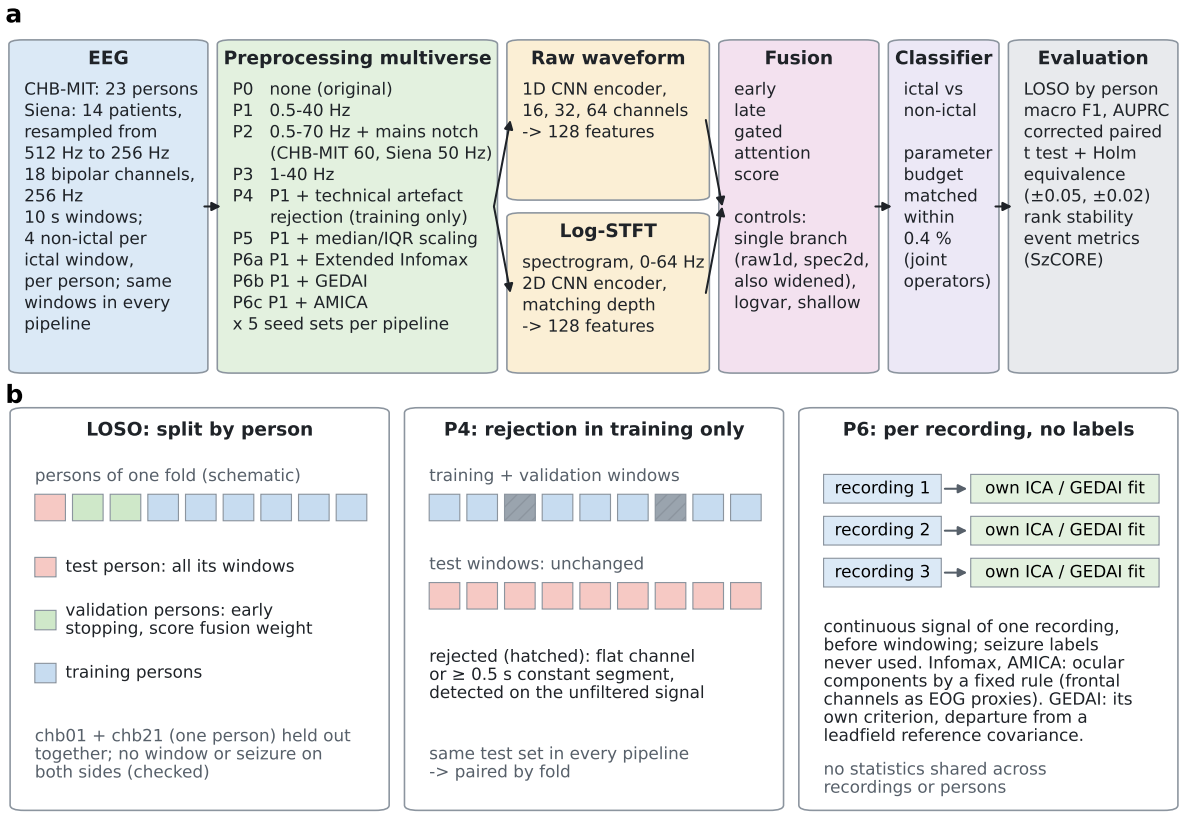


Figure 1: Study design. **a**, The multiverse: two datasets harmonised to 18 bipolar channels at 256 Hz and 10 s windows; nine preprocessing pipelines, each trained with five seed sets; a 1D CNN on the raw waveform and a 2D CNN on the log-STFT, combined by five fusion operators, four jointly trained under a matched parameter budget and score fusion as a decision-level ensemble of the two branches at their own size, with single-branch and classical controls; evaluation by leave-one-subject-out (LOSO) at the window and the seizure level. **b**, Where each step is applied and how leakage is prevented: whole persons are held out (chb01 and chb21 together), and validation persons serve only early stopping and the score fusion weight; P4 rejects windows from training and validation data only, so every pipeline is tested on the same windows; P6 cleans each recording on its own, without seizure labels.

per ictal window of that subject, with a fixed seed, and every subject's true recording duration is kept for the event metrics (Section 2.5). This gives 6,380 windows (1,276 ictal) for CHB-MIT and 1,640 windows (328 ictal) for Siena, exactly four non-ictal windows per ictal window in every subject. The CHB-MIT windows come from 670 of the 683 recordings; no window was drawn from the other 13, all of them without seizures. The same windows are used by every pipeline, which is checked whenever a pipeline's corpus is built, so all comparisons between pipelines are paired window by window.

Class prevalence. The fixed ratio gives every pipeline, operator and subject the same class distribution, so that window-level results are comparable across them. Consequently, the window-level evaluation reflects this controlled sampling distribution, with 20 percent ictal windows, rather than the natural prevalence of seizures in continuous EEG, which is far lower (0.4 percent of all 10 s windows of the CHB-MIT recordings, 0.6 percent in Siena's recordings, which are cut around seizures). Window-level metrics, and macro F1 at a fixed threshold in particular, should therefore be interpreted within this experimental setting and not as the performance expected on continuous recordings. The seizure-level evaluation (Section 2.5) runs on the original, complete recording timelines and complements it.

Differences between the datasets. The two corpora differ in age (mostly children against adults), seizure types, recording length per patient (Siena's recordings are cut around seizures for a seizure prediction study), centre, equipment and mains frequency (60 Hz in Boston, 50 Hz in Italy). Results are compared between them as an independent replication and a transfer test (RQ3), not interpreted as an age or population effect: because the datasets differ in several of these respects at once, no difference between them can be attributed to any single one.

2.2 Preprocessing multiverse

Following the multiverse approach (Steegen et al. 2016), nine defensible preprocessing pipelines are crossed with every model, operator and seed instead of choosing one. Each pipeline changes one hypothesis-driven aspect; windows, labels, spectrogram settings, models, training and evaluation are fixed (`src/preprocess.py`).

Pipeline	Band-pass	Notch	Artefact handling	Normalisation	Question
P0	none	none	none	z-score	original benchmark
P1	0.5 to 40 Hz	none	none	z-score	standard EEG band
P2	0.5 to 70 Hz	mains (CHB-MIT 60 Hz, Siena 50 Hz)	none	z-score	information above 40 Hz
P3	1 to 40 Hz	none	none	z-score	low frequency content and drift
P4	0.5 to 40 Hz	none	technical artefact rejection (training only)	z-score	artefact rejection
P5	0.5 to 40 Hz	none	none	median and IQR	normalisation
P6a	0.5 to 40 Hz	none	Extended Infomax ICA	z-score	artefact removal
P6b	0.5 to 40 Hz	none	GEDAI	z-score	artefact removal
P6c	0.5 to 40 Hz	none	AMICA	z-score	artefact removal

Order of operations. Each continuous recording is band-pass filtered, notch filtered (P2) and

cleaned (P6), in that order; it is then cut into the fixed windows, P4's rejection rule is applied, and each window is normalised. The spectrogram is computed from the normalised window.

Filters. Zero-phase Butterworth filters (order 4, forward and backward) are applied to each continuous recording before it is windowed, so that no filter transient falls at a window edge. P2's notch removes each dataset's own mains frequency and its first harmonic; with a fixed 60 Hz notch Siena's P2 windows kept a 50 Hz line 318 times stronger than the 40 to 48 Hz background, so P2 was defined per dataset and its Siena runs were repeated.

Normalisation. Each channel of each window is standardised on its own: z-score (P0 to P4, P6) or median and interquartile range (P5). No statistic is computed across windows, recordings or persons, so normalisation cannot carry information from test to training data.

P4, technical artefact rejection. Amplitude was measured before a criterion was chosen: every amplitude threshold removes seizures selectively (500 μV rejects 72 percent of CHB-MIT's ictal and 48 percent of its non-ictal windows), because ictal windows have larger amplitudes. P4 therefore rejects only technical artefacts: a flat channel (standard deviation below 1 μV in the window) or a constant run of at least 0.5 s on any channel (a signal dropout), detected on the unfiltered signal. In CHB-MIT this rejects 42 of 1,276 ictal (3.3 percent) and 39 of 5,104 non-ictal windows (0.8 percent); the difference reflects which recordings have dropouts, not physiology (Supplementary Table S3). In Siena it rejects one window. Rejected windows are removed from training and validation only; the test windows are those of every other pipeline.

P6, three artefact removal algorithms. P6 applies Extended Infomax ICA (Lee et al. 1999; MNE-Python, Gramfort et al. 2013), GEDAI (Ros et al. 2025) or AMICA (Palmer et al. 2008; `jamica`) to the same 0.5 to 40 Hz signal, recording by recording. Infomax and AMICA share every setting and one component rejection rule: 16 components (the 18-channel bipolar montage has rank 16, because it contains two closed electrode loops per hemisphere), fitted on every 8th sample, with components removed when their correlation with any of the four frontopolar channels, used as EOG proxies since neither dataset has an EOG channel, has $|z| > 3$ (MNE `find_bads_eog`). Muscle components are not removed, because the gamma band carries the strongest ictal signal in CHB-MIT. GEDAI is not an ICA: it removes what departs from a leadfield-based reference covariance, by its own criterion, so the shared rule cannot apply. Its reference covariance C, defined for a referential layout, is mapped onto the bipolar montage as $D\,C\,D^{\top}$ (D the electrode difference matrix), and the channel mean is removed before and restored after cleaning. GEDAI's threshold preset was fixed before any classification run, on signal preservation alone: on six CHB-MIT subjects the default removed more than half of the power of 39 percent of recordings, the conservative preset of 17 percent; the conservative preset removed no more from ictal than from non-ictal windows, whereas the default kept a lower median share of the ictal power (0.78 against 1.00; Supplementary Table S2); P6b uses the conservative preset. The comparison among P6a, P6b and P6c is therefore a comparison of whole cleaning procedures: for Infomax and AMICA it isolates the decomposition algorithm, while GEDAI differs also in how it decides what to remove. Signal retention and compute are in Supplementary Table S1, and the power removed by band and class in Supplementary Table S5.

No leakage through P6. Each recording is decomposed and cleaned on its own, from its own unlabelled signal; seizure annotations are never used, and no decomposition, threshold or statistic is shared between recordings or persons. A test person's recordings are therefore cleaned exactly as they would be without any training data.

Offline processing. Filtering and artefact cleaning are applied to complete recordings before windows are extracted, so a window's cleaned signal depends on data recorded after it. These experiments therefore evaluate the robustness of offline preprocessing and do not represent a causal, real-time seizure detection pipeline, in which only past signal is available.

2.3 EEG representations and fusion strategies

Representations. Each 10 s window enters the network twice: as the 18-channel waveform (2,560 samples per channel) and as its log-magnitude short-time Fourier transform, log(1 + |STFT|), computed per channel with a 1 s Hann window, 0.5 s hop and 256-point FFT and cropped at 64 Hz (65 frequency bins by 19 frames; `src/chbmit_prep.py`). The 64 Hz ceiling is kept in every pipeline, so input shapes and parameter counts never change.

Architecture. A 1D convolutional encoder (channel widths 16, 32, 64) processes the waveform and a 2D convolutional encoder of matching depth the spectrogram; each yields a 128-dimensional pooled feature vector. Five interchangeable fusion operators combine them:

- **Early:** the waveform passes a small strided 1D encoder, is broadcast along the spectrogram's time axis and concatenated to it as an extra channel before the 2D backbone (architecturally an intermediate fusion, since the two inputs share no tensor shape).
- **Late:** the two feature vectors are concatenated and classified by a small MLP.
- **Gated:** a learned scalar gate per branch reweights the concatenated features.
- **Attention:** a learned attention module weights the branches from both feature vectors jointly.
- **Score:** the two branches are trained as independent classifiers and their probabilities blended, p = w p_waveform + (1 - w) p_spectrogram, with the weight w chosen on the validation subjects only, by minimising their log loss over a 41-point grid from 0 to 1.

Parameter matching. Every jointly trained model (the four joint operators and two widened single-branch controls) is sized to a common budget by solving the fusion head's hidden width numerically (`src/models.py`): with 18 input channels the models have 102,236 to 103,712 parameters, the four joint operators within 0.36 percent of each other. Score fusion, an ensemble of two independently sized classifiers, keeps its natural size: it blends the probabilities of the two default-width single-branch controls below (28,850 and 34,546 parameters, 63,396 together, about 62 percent of the budget). Score fusion is retained as a complementary decision-level strategy: because its architecture and training differ from those of the jointly trained models, it is part of the broader practical comparison of operators but not of the strictly parameter-matched architectural comparison, which concerns early, late, gated and attention fusion (102,547; 102,768; 102,921 and 102,767 parameters).

Controls. The two single-branch networks at their default width (raw1d, spec2d) and widened to the fusion budget (raw1d_wide, spec2d_wide), which asks whether fusion beats a unimodal model of the same size; and two classical baselines without representation learning, a logistic regression on log variance (logvar) and on seven hand-crafted features (log variance, line length and relative power in five bands; shallow), 19 and 127 parameters with the intercept. The networks see the same windows and preprocessing as the operators. The logistic regressions (standardised features, class-balanced weights, C = 1, no tuning) need no early stopping, so they are fitted on all training subjects, including the four (Siena: three) that the networks set aside for validation; this favours the classical baselines, if anything.

2.4 Experimental protocol

Leave one subject out. All evaluation is subject-independent. Each subject is held out once as the test set (23 folds on CHB-MIT, 14 on Siena), and all recordings of one individual are assigned to the same partition (chb01 and chb21 are held out together). Within each outer fold, validation subjects are selected exclusively from the remaining training subjects, and the held-out subject is excluded from all training, early stopping and model selection. The selection is deterministic

and does not look at any result: the remaining subjects are ordered by their number of ictal windows, and round(0.2 n) of them (4 of 22 on CHB-MIT, 3 of 13 on Siena) are taken at evenly spaced positions of that order, so the validation set spans the range of seizure burdens and always contains ictal windows. It serves early stopping and the score fusion weight only. No subject's windows and no seizure's windows ever appear in two of the training, validation and test partitions; an automatic check of every fold confirms this for both datasets and that every stored run held out the same subjects in the same folds (`src/audit_methods.py`), and the leakage checker was itself validated on a deliberately leaky split (`src/chbmit_corpus.py`).

Training. AdamW (Loshchilov and Hutter 2019), learning rate 1e-3, weight decay 1e-4, batch size 32, class-weighted cross-entropy; early stopping on validation macro F1 with patience 12 after a minimum of 20 epochs, at most 60 epochs. The minimum guards against a model stopping in a trivial single-class solution, found in an early audit. These settings, and the architectures, were fixed before the multiverse runs and not tuned: there was no hyperparameter search or optimisation, the same values were used for every model, pipeline and dataset, and the validation subjects served only early stopping and the score fusion weight. The reported performance is therefore that of a fixed, untuned configuration rather than the best each operator could reach.

Seeds. Every pipeline is run with five seed sets. A run's seed for each fold and model is derived from the seed set, fold and model (`src/config.py: fold_seed`), so it does not depend on execution order. It sets the weight initialisation, the order of the mini-batches and the dropout masks. It does not change the data: the window sampling (one fixed seed), the validation subjects (deterministic) and the artefact cleaning (a seed per recording) are the same in every seed set, so differences between seed sets reflect stochastic training alone. The ranking of the operators was unstable across three seed sets (Kendall's W below 0.5 in seven of nine CHB-MIT pipelines); after these three-seed results, the extension to five seed sets was decided, to estimate the seed variance more reliably. It was not a rule fixed beforehand.

Cross-dataset experiment (RQ3). Each model is trained once on all CHB-MIT subjects (with the same inner validation rule) and tested on every Siena window and recording, for every pipeline and seed set (`src/cross_dataset.py`). The pipeline is applied to each dataset as defined, including each dataset's own mains notch in P2. Siena is used for testing only: the validation subjects are drawn from CHB-MIT, and no Siena window enters training, early stopping, the score fusion weight or any other choice. Results are reported per Siena patient. The experiment asks whether the relative performance of the operators persists when the dataset changes. Because the datasets differ in population, acquisition and recording protocol (Section 2.1), it measures their combined distribution shift and does not isolate any single source of it.

2.5 Evaluation metrics

Window level (primary). Macro F1 and the area under the precision-recall curve (AUPRC) are primary; balanced accuracy (the mean of sensitivity and specificity) is reported alongside, with AUROC and the Brier score as secondary. A window is predicted ictal when its ictal probability exceeds 0.5, for every model and control alike. AUPRC is computed as average precision (scikit-learn) from the predicted ictal probability, with the ictal class as the positive class; it assesses how well the probabilities rank ictal above non-ictal windows across all thresholds, complementing the threshold-dependent macro F1. Metrics are computed per fold on the held-out subject's windows and averaged over folds within a run, each fold with equal weight. The subject is the unit of generalisation and of the statistical tests; pooling windows instead would let the subjects with the most seizures dominate the average.

Window-level classification and seizure-level detection are evaluated separately because they address different aspects of performance: macro F1 and AUPRC describe classification under the controlled window sampling (Section 2.1), whereas seizure-level sensitivity and false alarms per

hour are assessed on the original recording timelines with event-scoring rules fixed in advance. A model can classify windows well and still miss seizures or raise many false alarms.

Seizure level (secondary, clinical). Every model saved for a fold is applied to every 10 s window of every recording of the held-out subject, not only the subsampled test windows, after the pipeline's own preprocessing; a window is positive when its probability exceeds 0.5, with no further post-processing. Detections are scored against the annotated seizures with SzCORE's event scoring (Dan et al. 2024; `timescoring` 0.0.7, defaults: 30 s tolerance before onset and 60 s after offset, any overlap counts, events closer than 90 s merged, events longer than 5 min split). Seizure-level sensitivity is the share of seizures detected; the false alarm rate is false-positive events per hour on the true recording time. The scored recordings hold 187 seizures in 979.9 h (CHB-MIT) and 47 seizures in 141.0 h (Siena). On CHB-MIT these are the 185 annotated seizures of the 683 recordings in the target montage (of 198 in the dataset, Section 2.1); one of them, in chb11_99, lasts 752 s and is split by the 5 min rule into three reference events, and in no other recording does merging or splitting change the count (`results_v2/audit/methods_audit.md`). Scoring the window labels themselves as predictions detects all of them without a false alarm, which checks the reference, the windows and the scoring together. On Siena only sensitivity is reported: its recordings are cut around seizures, so a false alarm rate would not describe continuous monitoring. The criterion, the merging of consecutive detections and the counting of false alarms were fixed, and committed, before any multiverse run (`docs/EVENTS.md`), and are the same for every operator and pipeline.

2.6 Statistical analysis

Pairwise comparisons. Leave-one-subject-out folds share training data, so the standard paired test overstates significance; in a simulation of two equivalent classifiers the naive paired Wilcoxon test rejected in 37.6 percent of cases at a nominal 5 percent (`src/sim_cv_correlation.py`). On the Bonn corpus, studied in an earlier stage of this project, the naive test found 13 to 22 significant pairs among 55 per task and the corrected test none among the deep models (Supplementary Note 1). Because LOSO training sets overlap substantially, pairs of operators are compared with the resampling-corrected paired t test (Nadeau and Bengio 2003) rather than treating the held-out folds as independent observations. The test inflates the variance of the mean fold difference from var(d)/k to (1/k + n_test/n_train) var(d), on k - 1 degrees of freedom. We count subjects: n_test/n_train = 1/22 on CHB-MIT and 1/13 on Siena, the held-out subject against the subjects outside it. Since 4 (Siena: 3) of these serve as validation subjects, the networks are fitted on 18 (10); with that conservative ratio, 1/18 and 1/10, no operator pair becomes significant and the pairs equivalent within 0.05 fall from 73 to 70 of 90 on CHB-MIT and from 21 to 19 on Siena (`results_v2/audit/methods_audit.md`). Subjects contribute different numbers of windows, but each fold enters the test as one observation (Section 2.5). Seed sets are averaged within each fold before testing, so the five seed sets and nine pipelines never inflate the number of independent observations, which is always the number of subjects.

Multiplicity is addressed with Holm correction within predefined families: the ten operator pairs of one pipeline, for one dataset and metric, form one family (nine families per dataset); the 30 comparisons of the six controls with the five operators in one pipeline form another.

Practical equivalence. For each pair the two one-sided test with the same corrected variance gives the smallest margin within which the pair is equivalent at alpha = 0.05; a pair is equivalent if that margin is at most 0.05 macro F1 (main analysis) and, as a stricter sensitivity analysis, 0.02. Both margins were fixed and committed before any multiverse result existed. A pair is robustly equivalent if it is equivalent in every pipeline. Statistical significance and practical equivalence are assessed separately: failure to reject equal performance is not interpreted as evidence of equivalence. The corrected test, the equivalence bound, the Holm adjustment and

Kendall's W were checked against independent re-implementations (`src/audit_methods.py`).

Ranking stability. In every run (pipeline and seed set) the five operators are ranked by mean macro F1. We report each operator's full rank distribution (counts per rank, median, best and worst rank, share of runs won and lost), Kendall's W across seed sets within each pipeline, and Kendall's tau between the seed-averaged rankings of every pair of pipelines. Whether preprocessing changes the ranking more than the seed does is tested by comparing the mean tau between seed sets of the same pipeline with the mean tau between pipelines, against a null distribution from 10,000 permutations of the pipeline labels within each seed set; the same test is run among P6a, P6b and P6c alone.

Fusion x Preprocessing. A repeated-measures ANOVA on macro F1 (seed means) treats the held-out subjects as the repeated units and pipeline and operator as within-subject factors; each effect is tested against its interaction with subject, with Greenhouse-Geisser correction. Effect sizes are reported as partial eta squared.

Variance components. For macro F1, AUPRC and balanced accuracy, the fully crossed subject x pipeline x operator x seed design is decomposed as y = mu + subject + pipeline + operator + seed + every interaction, all treated as random effects (generalizability theory) and estimated by the method of moments from the expected mean squares. With one observation per cell the four-way interaction is the residual. Every component is reported unrounded, as an estimated variance, as a standard deviation in the metric's own units (a practical size) and as a share of the total; negative estimates, which arise when an effect's mean square is smaller than its error term, are reported as such and set to zero in the shares. The components are descriptive estimates of the variability associated with the held-out subject, the pipeline, the operator and stochastic training. A share of zero is therefore not proof that an effect is absent, and the subject component is read as subject-associated variability: it collects everything that differs between held-out subjects, of which individual physiology is only one part, together with seizure count and type, recording length and recording conditions.

Prediction stability. The agreement (Cohen's kappa) of window-level predictions between pipelines at the same seed set is compared with the agreement between seed sets in the same pipeline.

Statistical significance is reported from the corrected tests and the ANOVA, practical size from the equivalence margins and the variance components in metric units, and ranking stability from the rank analyses; a result in one of these three is never used as evidence for another. All analyses are implemented in `src/multiverse.py` and were validated on synthetic runs with known effects (39 checks).

2.7 Reproducibility and computational cost

Software and hardware. Two environments were used (Table 2); artefact removal needs a newer numpy than the training code and runs in its own environment (`requirements.txt`, `requirements-ica.txt`). The software versions and the hardware of every run are recorded in its `meta.json`.

Table 2: Computational environments.

	Training and window-level evaluation	Artefact removal (P6) and seizure-level evaluation
Used for	all LOSO and cross-dataset runs (135 runs)	the P6 corpora; applying the saved models to every window of every recording

	Training and window-level evaluation	Artefact removal (P6) and seizure-level evaluation
Machine	lab PC: 2 x Intel Xeon (Skylake-SP), 64 logical CPUs; NVIDIA GeForce RTX 5070	laptop: Intel Core Ultra 7 155U, 14 logical CPUs; no CUDA GPU
Operating system	Windows 10 (build 19045)	Windows 11 (build 26300)
Python	3.12.10	3.12.7
Numerical libraries	PyTorch 2.8.0 (CUDA 12.8 build), numpy 1.26.4, scipy 1.16.1, scikit-learn 1.7.2	PyTorch 2.8.0 (CPU build), numpy 2.5.3 (OpenBLAS), scipy 1.16.1, scikit-learn 1.7.2, MNE 1.13.2, GEDAI 0.60, jamica 0.3.0, timescoring 0.0.7
Threads	16 CPU threads	14 (BLAS)
Seeds	per seed set, fold and model (`fold_seed`)	per recording, from its file name (crc32)
Determinism	PyTorch deterministic algorithms on the GPU	CPU only

Determinism and device. All training runs of one analysis use one machine, one device and 16 CPU threads, and CPU and GPU runs are never mixed in one analysis. On the CPU a run was reproduced bit for bit, also with two runs sharing the machine. The multiverse ran on the GPU with PyTorch's deterministic algorithms; adaptive average pooling has no deterministic CUDA backward, yet two GPU runs of the same fold were identical. Before switching to the GPU, its agreement with the CPU was tested at identical seeds: over five folds the mean absolute macro F1 difference between CPU and GPU was 0.074, slightly larger than the 0.066 between two CPU seed sets (`python -m src.compare_runs`; `docs/PIPELINE.md`). The criterion fixed beforehand, a CPU-GPU difference no larger than the difference between seed sets, was therefore narrowly missed. The CPU-GPU difference is comparable in size to the variation between stochastic training runs; this similarity does not establish that hardware-related numerical differences and seed effects arise from the same mechanism, and we make no claim about the cause.

Reproducibility conditions. Reproducibility was examined at the level of the preprocessing and of the downstream predictions (Table 3). Identical preprocessing configurations were run in different computational environments, and the differences in the cleaned signals were followed into the classification and seizure-level metrics.

Table 3: What was held fixed and what varied in each reproducibility comparison.

Comparison	Held fixed	Varied	Measured
Training, repeat on CPU	code, data, seeds, machine, threads	nothing (also two runs sharing the machine)	fold metrics: bit-identical
Training, repeat on GPU	code, data, seeds, machine, GPU, deterministic mode	nothing	fold metrics: identical
Training, CPU against GPU	code, data, seeds, machine	device (CPU or GPU)	macro F1 difference, against that between seed sets
Artefact removal, two machines	code, input recordings, package versions, seeds, BLAS kernel	machine (laptop, lab PC), thread count	components removed, power kept, per recording; 670 CHB-MIT recordings
Artefact removal, same machine	code, input, versions, seeds, machine	process parallelism (run alone or in parallel workers)	as above
Artefact removal, third machine	code, input, versions, seeds	architecture (ARM, macOS)	as above, on a subset of recordings

Comparison	Held fixed	Varied	Measured
Rounding perturbation	everything	input multiplied by 1 + 1e-7 N(0, 1)	as above
Effect on outcomes	trained models, test recordings	cleaned signals (training-time against event-time cleaning)	seizure-level sensitivity, false alarms, rankings

Reproducibility of artefact removal. Fixing seeds and package versions did not make P6 reproducible across machines and execution contexts (`docs/ICA_REPRO.md`). AMICA removed the same components in every recording on every machine and in every execution context; the power it kept differed by at most 0.003 between machines and 0.0007 between two runs on the same machine. Extended Infomax removed different components in recordings whose component lay near the $|z| = 3$ rule: 2.5 percent of recordings between two machines and 1.2 percent between two runs on the same machine with different process parallelism. GEDAI was deterministic on each machine but gave a different result on the second machine in 27 percent of recordings, often by a large amount. A third machine of another architecture (ARM) agreed with the other two on recordings where they agreed, and often with neither where they differed. Multiplying the input by (1 + 1e-7 N(0, 1)), a rounding-level perturbation, changed GEDAI's result on exactly the recordings that differed between machines, also on a full-rank montage, and Infomax's component decision on recordings where it had differed; AMICA was unchanged. This is consistent with a sensitivity of the methods to rounding near a decision threshold on particular recordings rather than with a fault of one machine or library, but it does not identify the numerical mechanism, and the differences are reported as an empirical reproducibility limitation. All three methods were compared on the same criteria: the components removed and the share of signal power kept, per recording. Window-level results are unaffected, because every P6 training and test window comes from one set of signals, cleaned once. The event metrics, which need every recording cleaned again, were computed on the machine that produced those signals; they then reproduce the training-time cleaning exactly for GEDAI, within 1.4 percent of the amplitude for AMICA, and differ for Infomax in 8 of 670 CHB-MIT recordings (`events/meta.json`, `window_check`).

Computational cost. Artefact removal of the 670 CHB-MIT recordings with selected windows took 4.7 (GEDAI), 8.5 (Infomax) and 73 (AMICA) CPU hours on a laptop CPU (Supplementary Table S1). One CHB-MIT run (23 folds, nine models) took about 1.5 h on the GPU, one Siena run about a quarter of an hour, and one cross-dataset run about 5 min; training one fold (all nine models) took 3.5 to 6 min on the GPU against 16 to 17 min on the CPU (`perfold.csv`).

Availability. Code, the per-fold results of every run, and all analyses are in the public repository, whose README lists the commands that regenerate each table and figure. Following our own recommendation (Section 4.5), the repository also holds, per recording and artefact removal method, the outcome of the cleaning the models were trained and tested on (components removed, GEDAI's threshold, power kept; `results_v2/qc/p6_recording_logs.csv`) and of the recleaning used for the event metrics (`p6_recording_logs_events.csv`), so that a reader can check a reproduction recording by recording.

3. Results

The evaluation comprised nine preprocessing pipelines, five seed sets and five fusion operators, together with single-branch neural-network and classical controls. Each pipeline-seed configuration was evaluated with leave-one-subject-out (LOSO) cross-validation on CHB-MIT (23 subjects) and Siena (14 subjects), giving 45 runs per dataset; 45 further runs evaluated transfer from CHB-MIT to Siena (`results_v2/multiverse/chbmit_cuda`, `results_v2/multiverse/siena_cuda`, `results_v2/cross`). Means over runs are given with the standard deviation over runs.

The results are presented in seven stages. We first examine whether fusion operators offer a meaningful performance advantage and whether it justifies their architectural complexity. We then distinguish statistical significance from practical equivalence, assess the stability of the operators' ranking across pipelines and seeds, and identify the principal sources of performance variability. Finally, we examine the effects and the reproducibility of artefact removal, the seizure-level detection, and whether the performance patterns generalise across datasets.

3.1 Fusion complexity does not consistently translate into superior performance

The five fusion operators reached broadly comparable window-level performance across the preprocessing multiverse, although their relative behaviour differed between datasets (Table 4, Figure 2).

On CHB-MIT, mean macro F1 over the 45 runs ranged from 0.673 ±0.017 for score fusion to 0.686 ±0.016 for gated fusion; early, late and attention fusion reached 0.677 ±0.013, 0.681 ±0.015 and 0.683 ±0.014. Within a pipeline, the best and the worst operator differed by 0.013 to 0.035 macro F1 (median 0.021), and each operator's mean varied little across pipelines (SD 0.007 to 0.008).

On Siena, the operators differed more. Score fusion reached the highest mean macro F1 (0.670 ±0.024), followed by late (0.640 ±0.025), gated (0.631 ±0.031), attention (0.625 ±0.023) and early fusion (0.617 ±0.021). Within a pipeline the best and the worst operator differed by 0.042 to 0.070 (median 0.053), and the variation across pipelines was larger than on CHB-MIT (SD 0.011 to 0.024). The separation between operators was therefore dataset-dependent, and its size has to be weighed against its uncertainty (Section 3.2) rather than read as superiority from mean performance alone.

Increasing architectural complexity did not consistently improve performance. The spectrogram network widened to the fusion budget reached a mean macro F1 of 0.683 on CHB-MIT, within the range of the jointly trained operators (0.677 to 0.686), and 0.663 on Siena, above every jointly trained operator (0.617 to 0.640) and below only score fusion (0.670); it exceeded the best operator of a run in 9 (CHB-MIT) and 16 (Siena) of 45 runs. The waveform network was weaker (0.662 and 0.612 at the fusion budget, 0.663 and 0.609 at default width), and widening either branch to the fusion budget changed little (Supplementary Table S7). The shallow classifier, a logistic regression with 127 parameters on seven hand-crafted features per channel, was equally competitive: its mean macro F1 was 0.690 on CHB-MIT, above the best operator of a run in 12 of 45 runs, and 0.720 on Siena, above the best operator in 43 of 45 runs and in all nine pipelines. This advantage depends on the decision threshold on CHB-MIT, where the shallow classifier's macro F1 is the lowest of all models at 0.3 (0.642) and the highest at 0.7 (0.715), but not on Siena, where it is the highest at 0.3, 0.5 and 0.7 (`results_v2/sensitivity_preds/summary.md`).

These descriptive advantages did not amount to statistically significant superiority. Tested like the operator pairs (corrected test, Holm across the 30 control-operator comparisons of a pipeline; `src/controls_stats.py`), no control differed significantly from any operator in any pipeline of either dataset, in macro F1 or AUPRC; the smallest corrected p was 0.12 (shallow against early fusion on Siena), and equivalence within 0.05 macro F1 could be shown in 34 of 270 comparisons on CHB-MIT and 13 of 270 on Siena (Figure 2c, f). The controls therefore stood to the operators as the operators stood to each other, and the additional complexity of multimodal fusion was not accompanied by a reliable improvement in macro F1 under the conditions evaluated.

Classification at a threshold and the ranking of windows by probability behaved differently. On CHB-MIT gated fusion had the highest mean AUPRC (0.666 ±0.018) and early fusion the lowest (0.613 ±0.024); on Siena early fusion reached 0.496 ±0.035 against 0.631 ±0.033 for score fusion. Early fusion had the lowest AUPRC in all nine pipelines of both datasets (0.564 to 0.632 on

CHB-MIT against at least 0.631 for every other operator, 0.453 to 0.540 on Siena against at least 0.522), although its macro F1 at the 0.5 threshold differed much less. On CHB-MIT the fused models ranked windows better on average than a single branch (late, gated, attention and score fusion 0.655 to 0.666 against at most 0.618); on Siena the difference from the best single branch (0.581) was inconsistent, from -0.085 (early fusion) to +0.050 (score fusion). Score fusion, which blends the two default-width single-branch networks, had the lowest Brier score of all models (0.265 and 0.304), and early fusion the highest among the operators on CHB-MIT (0.330 against 0.294 to 0.300). Fusion complexity, thresholded classification and probability ranking are therefore distinct properties: an operator can reach a competitive macro F1 without discriminating equally well across thresholds.

Table 4: Window-level performance of the fusion operators and the main controls, mean and SD over the 45 runs of each dataset (fold means within a run). Per pipeline: Figure 2 and Supplementary Table S6; all six controls with their parameter counts: Supplementary Table S7.

Model	CHB-MIT macro F1	CHB-MIT AUPRC	CHB-MIT bal. acc.	Siena macro F1	Siena AUPRC	Siena bal. acc.
early	0.677 ±0.013	0.613 ±0.024	0.682 ±0.011	0.617 ±0.021	0.496 ±0.035	0.642 ±0.022
late	0.681 ±0.015	0.656 ±0.019	0.679 ±0.012	0.640 ±0.025	0.599 ±0.040	0.679 ±0.018
gated	0.686 ±0.016	0.666 ±0.018	0.683 ±0.013	0.631 ±0.031	0.591 ±0.034	0.674 ±0.022
attention	0.683 ±0.014	0.655 ±0.020	0.682 ±0.013	0.625 ±0.023	0.580 ±0.034	0.667 ±0.020
score	0.673 ±0.017	0.655 ±0.017	0.678 ±0.015	0.670 ±0.024	0.631 ±0.033	0.689 ±0.022
raw1d_wide (control)	0.662 ±0.012	0.605 ±0.017	0.673 ±0.010	0.612 ±0.028	0.577 ±0.033	0.657 ±0.019
spec2d_wide (control)	0.683 ±0.014	0.618 ±0.016	0.680 ±0.013	0.663 ±0.026	0.581 ±0.027	0.681 ±0.021
shallow (control)	0.690 ±0.008	0.640 ±0.008	0.739 ±0.007	0.720 ±0.019	0.667 ±0.015	0.770 ±0.017

3.2 Small performance differences do not establish practical equivalence

The narrow differences between operators called for a formal distinction between statistical significance and practical equivalence (Figure 3).

In both datasets, none of the 90 pairwise operator comparisons (10 pairs in each of nine pipelines) was significant after Holm correction; the smallest corrected p was 0.19 on CHB-MIT and 0.095 on Siena. The absence of a significant difference, however, did not mean that the operators were practically equivalent.

At the margin of ±0.05 macro F1 fixed in advance, 73 of the 90 comparisons on CHB-MIT were equivalent, and two pairs, late-attention and gated-attention, were equivalent in all nine pipelines. At the stricter margin of ±0.02 only one comparison was. On Siena the evidence for equivalence was much weaker: 21 of 90 comparisons were equivalent within ±0.05, no pair in every pipeline, and none within ±0.02. The smallest margin at which equivalence could be shown had a median of 0.040 on CHB-MIT (0.020 to 0.066) and 0.081 on Siena (0.021 to 0.148).

The two datasets therefore supported different inferences. CHB-MIT showed that many operators differ by less than a moderate margin, whereas Siena often lacked the precision to establish either a difference or equivalence. The results thus fall into three kinds: statistically detectable differences (none here), demonstrated practical equivalence, and inconclusive comparisons; treating every non-significant result as equivalence would have hidden this distinction. Supplementary Table S8 gives every paired difference with its confidence interval, corrected p and equivalence margin.

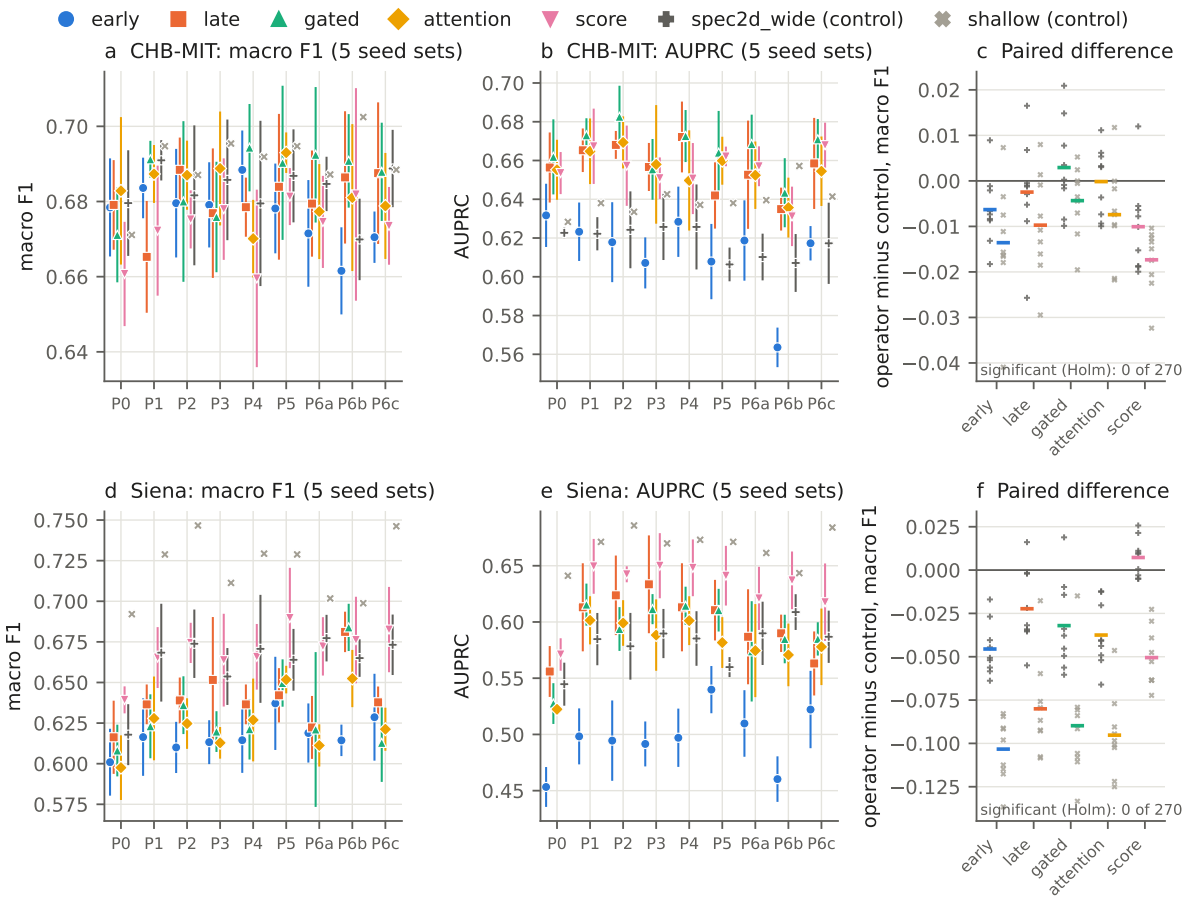


Figure 2: Fusion operators against the controls. **a, b, d, e**, Macro F1 and AUPRC of each operator and of two controls (the spectrogram network of the fusion budget, spec2d_wide, and the shallow logistic regression) in each preprocessing pipeline, on CHB-MIT (a, b; 23 LOSO folds) and Siena (d, e; 14 folds); mean and SD over five seed sets, each run averaged over its folds. **c, f**, Paired difference in macro F1 between each operator and each control, one point per pipeline (fold values averaged over seed sets), with the mean as a bar; none of the 270 control-operator comparisons per dataset was significant after Holm correction.

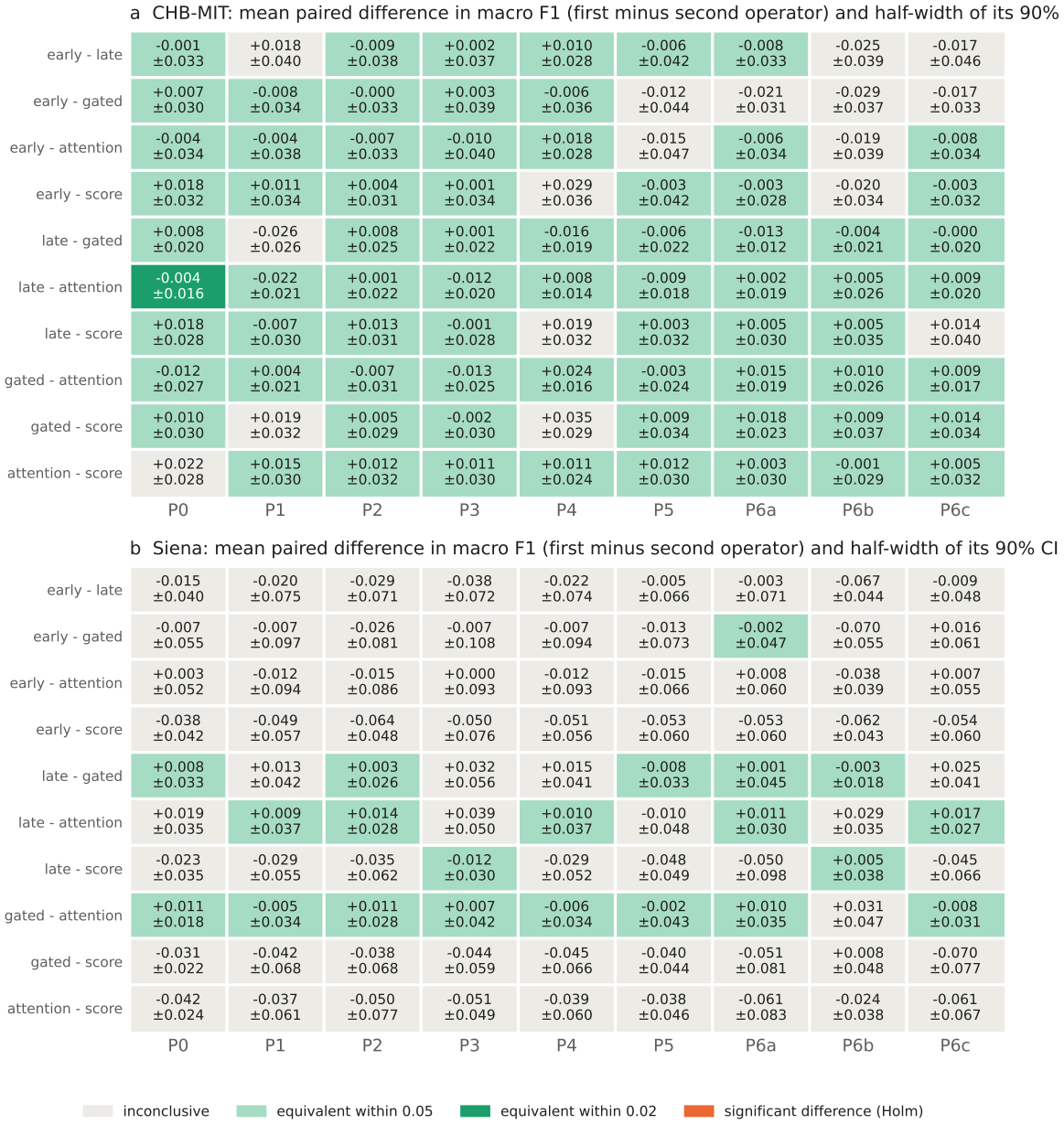


Figure 3: Equivalence matrix. For every operator pair (rows) in every preprocessing pipeline (columns): the mean paired difference in macro F1 (first minus second operator, fold values averaged over five seed sets) and the half-width of its 90 percent confidence interval with the corrected variance; the colour gives the outcome of the two one-sided tests at the margins of ± 0.05 and ± 0.02 and of the Holm-corrected test of a difference. **a**, CHB-MIT. **b**, Siena.

3.3 Operator rankings are sensitive to the seed and to preprocessing decisions

Although the mean differences were modest, which operator ranked first varied substantially between runs (Figure 4).

On CHB-MIT gated fusion ranked first in 38 percent of runs, followed by attention (24), late (20), early (11) and score fusion (7), and every operator ranked last in at least 13 percent of runs; no operator kept a consistently superior position. Changing only the seed reordered the operators almost completely: Kendall's W across the five seed sets of one pipeline had a median of 0.22 (0.14 to 0.62 over pipelines), and rankings from different seeds of the same pipeline agreed with a mean Kendall tau of 0.10. Rankings from different pipelines agreed with a mean tau of 0.03; the difference did not exceed what permuting the pipeline labels produces ($p = 0.073$), and among the three artefact removal pipelines alone there was no indication that changing the algorithm reordered the operators more than changing the seed (difference -0.07, $p = 0.88$). The window-level predictions changed about as much between pipelines as between seeds (Cohen's kappa 0.56 and 0.58).

On Siena the rankings were more consistent. Score fusion ranked first in 82 percent of runs and never last (mean rank 1.29); Kendall's W had a median of 0.58 (0.40 to 0.78), and rankings agreed with a mean tau of 0.37 between seeds and 0.34 between pipelines ($p = 0.074$). This consistency did not establish significant superiority: score fusion's advantage over the next operator, 0.030 macro F1 on average, lies within the range that fourteen subjects can neither confirm nor exclude, and the operator main effect was $F = 2.87$ (uncorrected $p = 0.032$, Greenhouse-Geisser $p = 0.092$, partial eta squared 0.18). An exception emerged among the three artefact removal pipelines: on Siena, changing the cleaning algorithm reordered the operators more than changing the seed (difference in mean Kendall tau 0.18, $p = 0.029$). The relative importance of preprocessing and of stochastic training therefore depends on the dataset and on the preprocessing decision compared.

The sensitivity of statistical inference to the number of seeds is illustrated by the Fusion $\times$ Preprocessing interaction on CHB-MIT. With three seed sets the Greenhouse-Geisser-corrected test gave $p = 0.034$; with five it gave $p = 0.069$ ($F = 1.73$, uncorrected $p = 0.008$, epsilon 0.34; partial eta squared 0.073). This change should not be read as an interaction that disappeared, but as a demonstration that conclusions drawn from a few training runs can depend on which seeds were included. On Siena there was no interaction with either number of seed sets ($p =$ 0.36 and 0.39).

3.4 Subject heterogeneity dominates the variability due to model and preprocessing choices

To identify the factors behind the variability, each metric was decomposed into subject, operator, preprocessing, seed and interaction components (Figure 5a, b).

On CHB-MIT the held-out subject accounted for 74.2 percent of the variance of macro F1 (SD 0.126), its interactions with pipeline and operator for 7.2 percent and the residual for 17.3 percent. All seed terms together accounted for 1.1 percent (SD 0.015), more than the operator and preprocessing, whose estimates were below zero (-2.4e-5 and -2.7e-5), that is, below their error terms (a zero share means that these effects could not be separated from their error terms, not that they are absent); the Fusion $\times$ Preprocessing component was 0.09 percent, consistent with the small practical size of the interaction despite its sensitivity to the number of seeds. AUPRC (subject 76.8 percent, operator 0.8, seed 1.5) and balanced accuracy (72.9, 0, 1.4) showed the same structure. Preprocessing had no detectable main effect in the repeated-measures ANOVA either ($F = 0.29$, Greenhouse-Geisser $p = 0.83$, partial eta squared 0.013).

On Siena the variability was more distributed. The subject accounted for 32.5 percent and its interactions with pipeline and operator for a further 26.9 percent; the operator held 1.5 percent

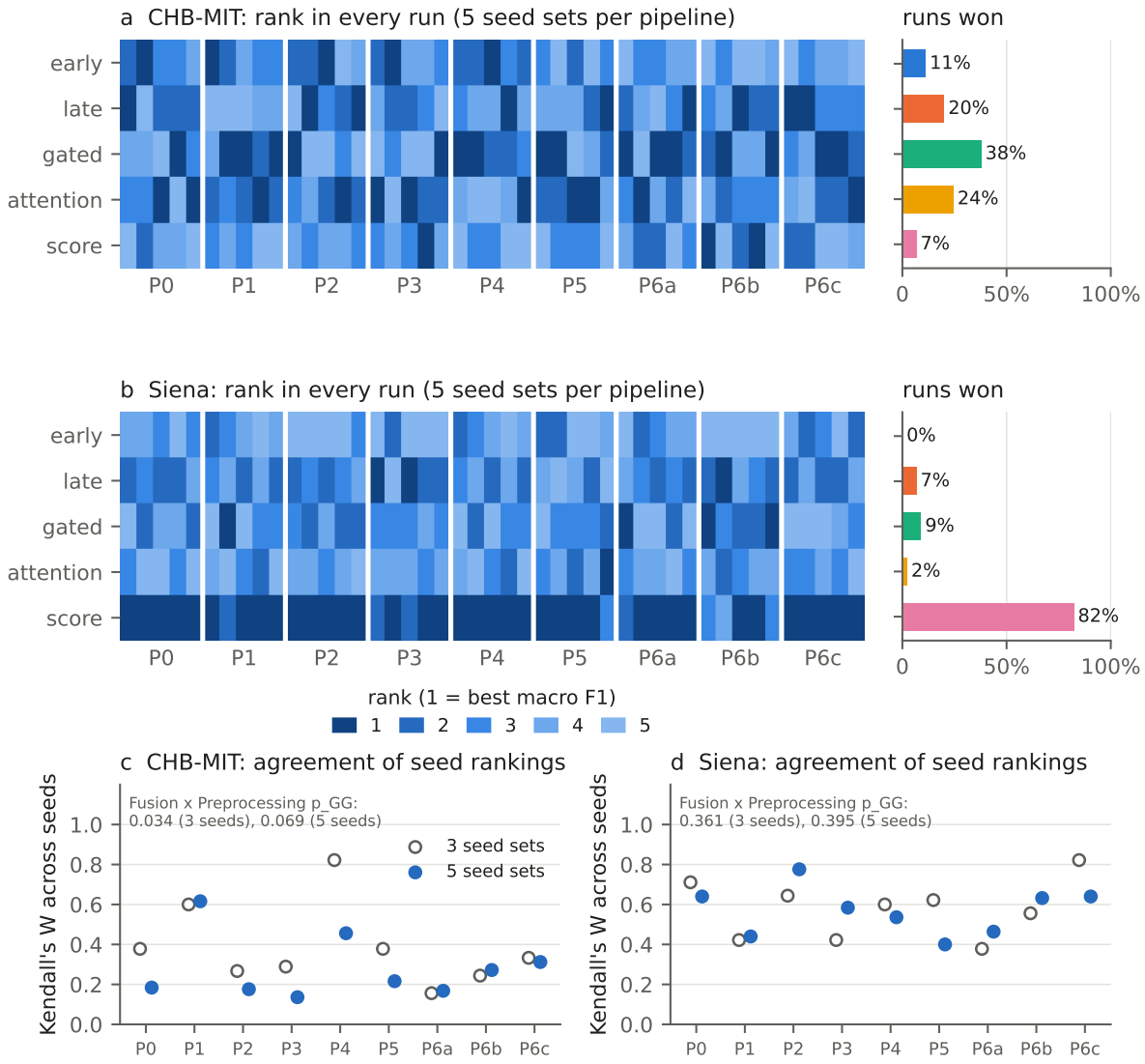


Figure 4: Ranking stability. **a, b**, Rank of each operator by macro F1 in every run (within each pipeline, one column per seed set; darker is better), and the share of the 45 runs of each dataset in which the operator ranked first. **c, d**, Agreement of the operator rankings across seed sets within each pipeline (Kendall's W) with three and with five seed sets, and the Greenhouse-Geisser-corrected p of the Fusion × Preprocessing interaction with each.

of the variance of macro F1 (SD 0.016) and 5.5 percent of AUPRC, the seed 4.2 percent and preprocessing 0.06 percent (main effect $F = 1.09$, $p = 0.36$).

The effects of fusion and preprocessing therefore cannot be characterised by their dataset-level averages alone. Mean macro F1 per held-out subject ranged from 0.46 (chb05) to 0.87 (chb22) on CHB-MIT and from 0.51 (PN07) to 0.78 (PN00) on Siena (Figure 5c, d). Within a subject the operators differed by a median of 0.063 macro F1 on CHB-MIT and 0.10 on Siena, three and two times the median difference within a pipeline, and the best operator changed from subject to subject: on CHB-MIT each of the five operators was the best for at least two subjects. An operator that suits one subject often does not suit another. This heterogeneity also explains why small differences in mean macro F1 were difficult to establish statistically: with 23 subjects on CHB-MIT and 14 on Siena, the uncertainty of every comparison is dominated by the variation between held-out subjects. The variance decomposition thus identifies subject heterogeneity as a central feature of the benchmark rather than as residual noise.

3.5 Artefact removal involves trade-offs between signal preservation, performance, cost and reproducibility

The three artefact removal procedures (Extended Infomax, GEDAI and AMICA) were evaluated for their effect on classification and also for signal preservation, computational cost and reproducibility (Figure 6).

Infomax and AMICA removed a median of one component per recording and kept a median of 78 percent of the signal's power; GEDAI kept 99.9 percent in most recordings but removed more than half of the power in 21 percent of them. They differed strongly in cost: cleaning the 670 CHB-MIT recordings took 8.5 CPU hours with Infomax, 4.7 with GEDAI and 73 with AMICA. Infomax and AMICA removed mostly delta power and, within a recording, slightly less from ictal than from non-ictal windows (a median difference of 0.010 and 0.017 of the power kept, Wilcoxon $p < 0.001$). Over the whole corpus, GEDAI's default preset kept a median of 63 percent of the power of CHB-MIT's ictal windows against 93 percent of the non-ictal ones, because the recordings it cleaned heavily held many of the seizures; within recordings it showed no difference (Supplementary Table S5).

The effect on classification was not uniform across operators or datasets. Against P1, the same signal without artefact removal, late fusion gained 0.014 to 0.022 macro F1 on CHB-MIT under all three algorithms and early fusion lost 0.012 to 0.022; on Siena GEDAI raised late and gated fusion by 0.045 and 0.061. AUPRC fell under every algorithm for most operators, by up to 0.060 (early fusion under GEDAI, CHB-MIT). No algorithm was better than the others for every operator and setting.

A separate analysis showed that the procedures differed in reproducibility (Section 2.7). With the same code, package versions and seeds, GEDAI cleaned about 27 percent of recordings differently on a second machine. Extended Infomax removed different components in about 2.5 percent of recordings between machines and in about 1.2 percent between two runs on the same machine with different process parallelism. AMICA removed the same components in every recording on both machines, and its retained power differed by at most 0.003. Experiments with a third machine of another architecture and with rounding-level perturbations of the input suggest that some cleaning decisions are sensitive to floating-point variation near the algorithms' decision thresholds.

These differences were not confined to intermediate signals. Scoring the same trained models on recordings cleaned on two machines changed seizure-level sensitivity by up to 0.054 under GEDAI and changed the order of the operators by sensitivity (Kendall tau 0.60 to 1.00); the change was at most 0.005 under Infomax and absent under AMICA. Reproducibility at the level

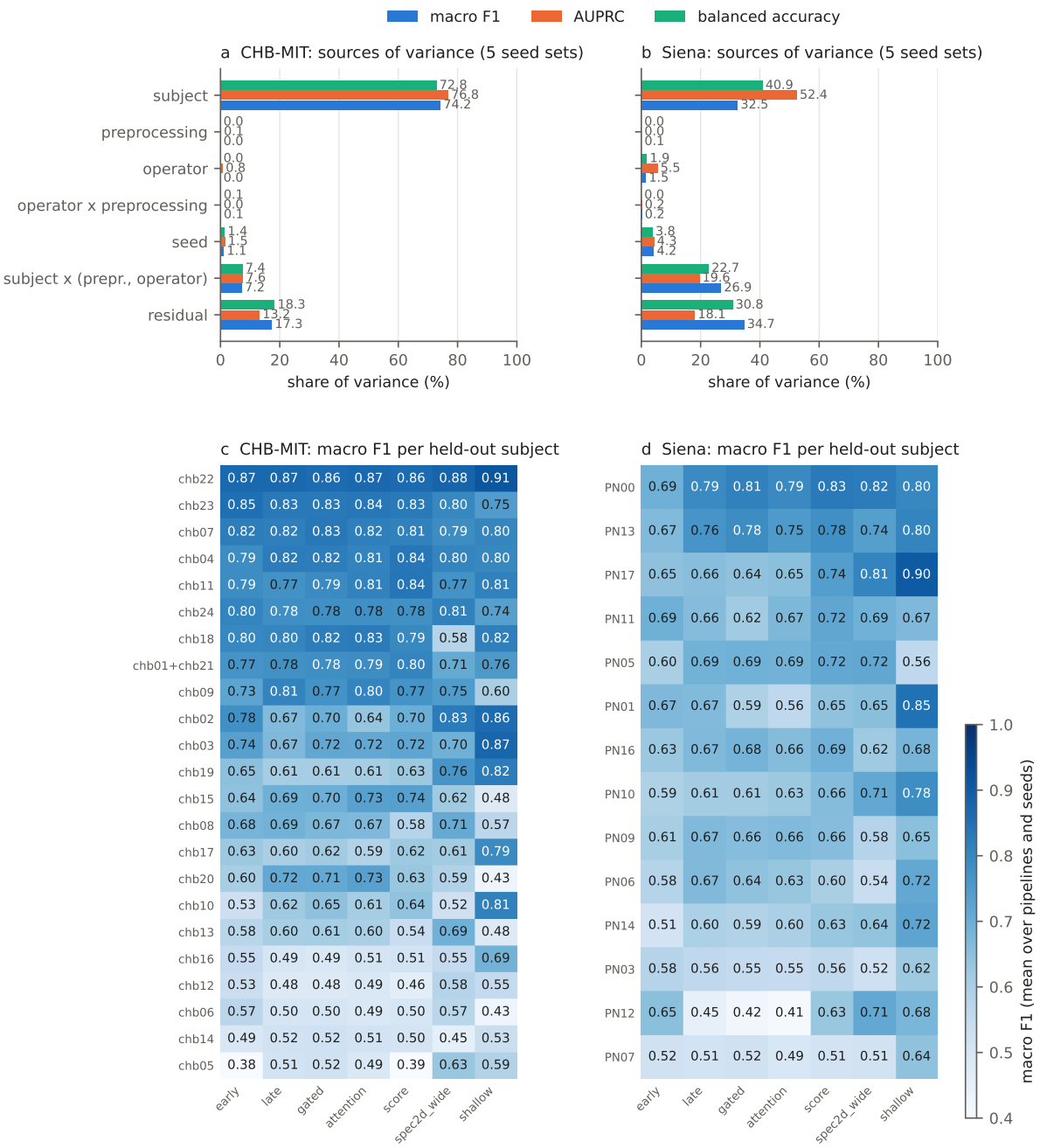


Figure 5: Sources of variability. **a, b**, Shares of the variance of macro F1, AUPRC and balanced accuracy (method-of-moments estimates for a fully crossed random-effects model of subject, preprocessing, operator and seed; negative estimates shown as 0; seed: all seed terms together). **c, d**, Macro F1 per held-out subject and model, mean over the nine pipelines and five seed sets; subjects sorted by their mean over the models.

of code, package versions and seeds therefore did not guarantee reproducible cleaned signals or downstream predictions. The results identify a trade-off: an inexpensive cleaning method can introduce more variation between machines, whereas a costlier one can give more stable outputs, and the reproducibility of preprocessing can affect the conclusions of a model comparison.

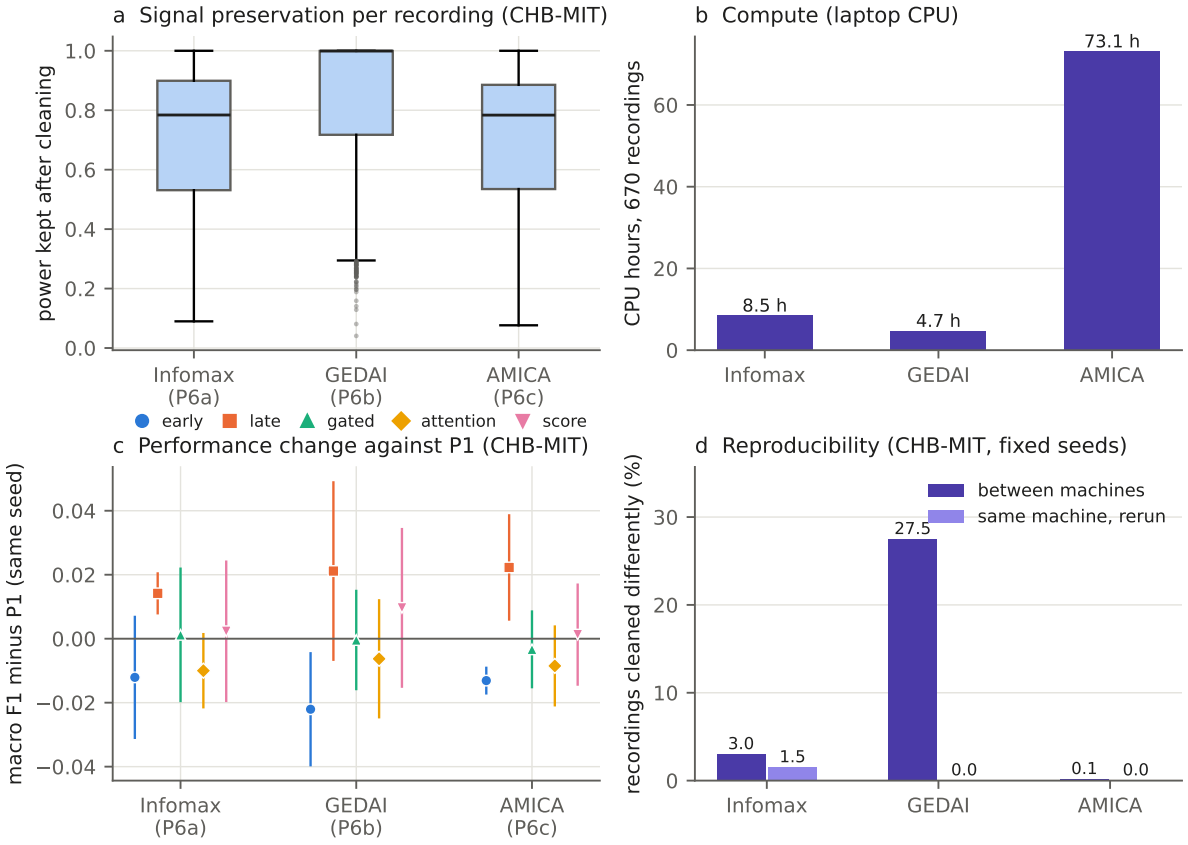


Figure 6: Artefact removal in P6 (CHB-MIT). **a**, Share of each recording's power kept after cleaning (670 recordings). **b**, CPU hours for the 670 recordings on a laptop CPU. **c**, Change in macro F1 against P1, the same 0.5 to 40 Hz signal without artefact removal, paired by seed set; mean and SD over five seed sets. **d**, Share of recordings cleaned differently (different components removed, or retained power differing by more than 0.001) on a second machine and in a second run on the same machine, with the same code, package versions and seeds.

3.6 Window-level differences translate into different seizure-level operating points

The window-level analysis was complemented by seizure-level evaluation on the complete recordings, with the probability threshold of 0.5 fixed in advance and without post-processing (Figure 7).

On CHB-MIT the operators detected 68 to 79 percent of the annotated seizures, with 5.2 to 7.7 false alarms per hour (`results_v2/results/events_chbmit.csv`). Early fusion had a distinct operating point: it reached the highest mean seizure-level sensitivity (0.79) and ranked first in sensitivity in 82 percent of runs, but it also raised the most false alarms in every run (7.7 per hour against 5.2 to 5.7 for the other operators). Taken together with the window-level results, in which early fusion had the lowest AUPRC in every pipeline of both datasets, this shows that a higher sensitivity at a given threshold does not imply better discrimination: early fusion labels more windows ictal, which detects more seizures at the price of more false alarms.

Artefact removal did not remove this trade-off. Under Infomax and AMICA the operators detected 66 to 76 percent of seizures with 4.8 to 7.2 false alarms per hour; under GEDAI, 71 to 80 percent with 5.5 to 8.7. On Siena the operators detected 84 to 87 percent of seizures, without a consistent order; false alarms per hour are not reported for Siena because its recordings are cut around the seizures and do not represent continuous monitoring.

Small differences in window-level macro F1 can therefore coexist with clearly different balances between seizure sensitivity and false-alarm burden, and window-level classification metrics alone do not characterise how a seizure detector would operate.

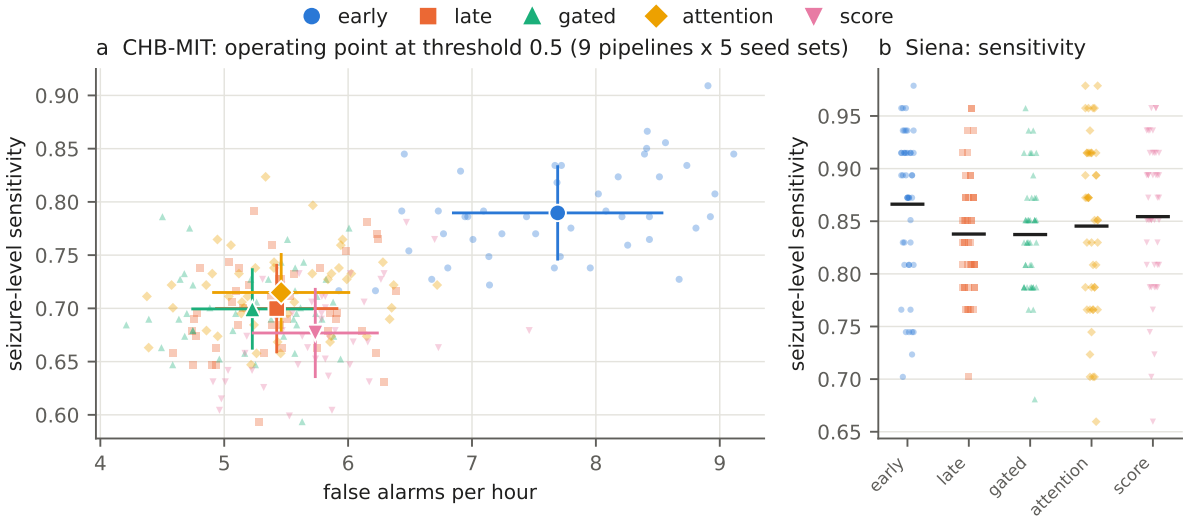


Figure 7: Seizure-level detection at the fixed threshold of 0.5 without post-processing. **a**, CHB-MIT: sensitivity against false alarms per hour, one point per run (nine pipelines × five seed sets) and the operator's mean and SD. **b**, Siena: seizure-level sensitivity per operator, one point per run, with the mean as a bar; false alarms per hour are not reported for Siena.

3.7 Predictive performance transfers across datasets, but operator rankings do not

Finally, we examined whether the patterns observed in subject-independent evaluation generalise across datasets (Figure 8).

Models trained on all of CHB-MIT and tested on Siena reached a mean macro F1 of 0.641 (attention) to 0.677 (late) over pipelines and seed sets. For early, late, gated and attention fusion this was higher than the mean obtained by training and testing on Siena (LOSO); score fusion was the exception, with 0.652 after transfer against 0.670 within Siena.

Comparable aggregate performance did not mean that the order of the operators carried over. Score fusion, first in 82 percent of Siena's own runs, was first in 11 percent of the transfer runs; there, late fusion ranked first most often (38 percent) and attention fusion was most often last (40 percent). Kendall tau between the seed-averaged operator rankings of the same pipeline was -0.11 on average between CHB-MIT and Siena, 0.07 between CHB-MIT and the transfer, and -0.11 between Siena and the transfer.

These results distinguish two forms of generalisation: whether a trained model keeps useful performance on an independent dataset, and whether the advantage of one fusion operator over another is preserved. The former was observed at the level of aggregate macro F1, the latter was not; an operator that ranked highly when trained and evaluated on one dataset did not necessarily keep that position under external evaluation.

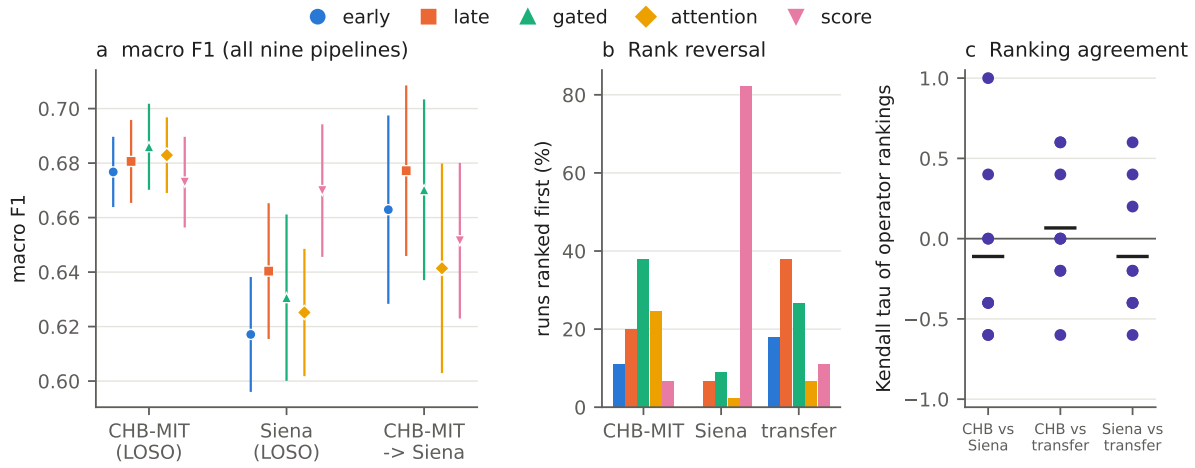


Figure 8: Cross-dataset generalisation. **a**, Macro F1 of each operator on CHB-MIT (LOSO), on Siena (LOSO), and trained on CHB-MIT and tested on Siena; mean and SD over the runs of all nine pipelines and five seed sets. **b**, Share of runs in which each operator ranks first in each setting. **c**, Kendall tau between the seed-averaged operator rankings of the same pipeline in two settings; one point per pipeline (points can overlap), bar: mean.

4. Discussion

We asked whether conclusions about fusion operators for EEG seizure detection hold when the preprocessing, the artefact removal algorithm, the training seed and the dataset change. On both datasets the answer is largely no: the operators performed within a few hundredths of macro F1 of each other, no difference between them was significant, and their order changed with the seed, with the artefact removal algorithm on Siena, and between datasets. We take the advisor's questions in turn.

4.1 How large are the differences between fusion operators?

Small. Within a pipeline the best and worst operator differed by a median of 0.021 macro F1 on CHB-MIT and 0.053 on Siena, and no pairwise difference survived the corrected test. The operator's share of the variance was estimated below zero on CHB-MIT and at 1.5 percent on Siena. One operator did stand apart in one respect: early fusion ranked windows by ictal probability worst in every pipeline of both datasets (lowest AUPRC in 18 of 18 cases), while its thresholded macro F1 was unremarkable and its seizure-level sensitivity, at the price of more false alarms, the highest. Early fusion is the only operator that combines the two representations before either branch has learned its own features, and its weaker ranking of windows is the one property of an operator that the multiverse preserved. Whether two operators are *equivalent* is a separate question, and the answer depends on the dataset: on CHB-MIT most pairs were equivalent within 0.05 macro F1 in most pipelines and two in all, but within 0.02 almost none, and on Siena few were equivalent even within 0.05. The data therefore rule out large differences between operators on CHB-MIT, but not differences of the size that published comparisons typically report as improvements.

The controls put these differences in proportion (Section 3.1, Table 4, Figure 2). A single-branch spectrogram network of the same size had a mean macro F1 similar to that of the jointly trained operators on CHB-MIT and higher on Siena, and a logistic regression on seven hand-crafted features had a mean macro F1 as high as or higher than every network, though not a higher AUPRC on CHB-MIT. Under the corrected test none of these differences was significant, nor

was the higher mean AUPRC of the fused models on CHB-MIT, and few were demonstrably equivalent: the controls stand to the operators as the operators stand to each other. Fusing the two representations may help the ranking of windows, by about as much as the operators differed from each other on CHB-MIT, but neither the choice to fuse nor the choice of operator changed performance as much as the subject did. This agrees with a broader evaluation of multimodal models, available as a preprint, in which added fusion complexity often did not pay off (Rheude et al. 2025). An improvement of one fusion operator over another says little without a parameter-matched unimodal control and a simple classical baseline evaluated in the same way.

4.2 Are the differences preserved when preprocessing changes?

Preprocessing had no detectable main effect on performance and did not reorder the operators more than a change of seed did. The interaction between operator and preprocessing, significant with three seed sets on CHB-MIT, was no longer significant after correction with five ($p = 0.069$), and its variance component was 0.09 percent. The one place where preprocessing mattered more than the seed was the choice among the three artefact removal algorithms on Siena ($p = 0.029$), where GEDAI raised late and gated fusion by about 0.05 macro F1. This agrees with the wider literature, where preprocessing changes results but no pipeline is best throughout (Section 1.1), and it adds that, for a comparison of operators, the pipeline mostly adds noise rather than a systematic advantage for one operator.

4.3 How much does the random seed change the rankings?

More than preprocessing does. On CHB-MIT the rankings from five seeds of the same pipeline agreed barely above chance (median Kendall's W 0.22), all seed terms together explained more of the variance than preprocessing and the operator, and the window-level predictions changed as much between seeds as between pipelines. A ranking obtained from one training run is therefore close to a random draw on CHB-MIT. The consequence for the literature is direct: a comparison of operators that reports one run per operator, as most do, cannot distinguish a better operator from a lucky seed at the differences observed here. Five seeds were enough to show this, and gave a more reliable estimate of the Fusion x Preprocessing interaction than three: significant with three seed sets, it was not with five (Section 3.3). That the seed explains more variance than the pipeline, shown with five seed sets, is a direct caution for the many comparisons that report a single run.

4.4 Why does the subject dominate?

Subject-associated variability made up 74 percent of the variance on CHB-MIT and, together with its interactions with pipeline and operator, about 60 percent on Siena. Seizures differ between people in morphology, location and duration, and the recordings in their equipment, montage quality and artefacts; a model trained on the other subjects works well or badly on a given person largely for reasons that no fusion operator or preprocessing step changes. The interaction of subject with pipeline and operator, 7 percent on CHB-MIT and 27 percent on Siena, means that the operator that suits one person is often not the one that suits another. Averaging over subjects hides this heterogeneity, and with 23 and 14 subjects it also limits the precision of every comparison between operators: the equivalence bounds of Section 3.2 are as wide as they are because the subjects differ so much.

4.5 Does the choice of artefact removal algorithm affect reliability?

In two ways. The three algorithms changed performance differently by operator (Section 3.5), and on Siena they reordered the operators more than the seed did. Less expected, they differed in reproducibility. With the same code, package versions and seeds, GEDAI cleaned about a quarter of recordings differently on a second machine, and Infomax changed its component decision in 2.5 percent of recordings between machines and in about 1 percent between two runs on one machine; scoring the same trained models on the two cleanings changed seizure-level sensitivity by up to 0.054 and the order of the operators. Fixed seeds and pinned versions are the reproducibility measures a methods section usually reports; for ICA-based cleaning they were not sufficient here, plausibly because decisions made at a threshold (a component's |z| against an EOG proxy, GEDAI's noise threshold) amplify rounding differences between machines. A perturbation of the input of one part in ten million produced differences of the same kind, which suggests that no machine gives the correct result and that choosing a different machine would not avoid them. The window-level results of this study rest on one cleaning and are unaffected; but a study that cleans its training data on one machine and its test data on another, or that compares its numbers with another group's, can see differences of the size of the effect it studies. Neuroimaging pipelines, including ICA decompositions of fMRI data, are known to give different results on different operating systems (Glatard et al. 2015); we are not aware of a comparable report for EEG artefact removal, where algorithms that take threshold decisions near floating-point precision can be deterministic within a machine (GEDAI) or even within one execution context (Infomax) and still not reproducible across machines. We recommend that studies using artefact removal store the cleaned signals, or at least the decomposition and the removed components, alongside the code.

4.6 Do results from one dataset carry over to another?

Performance did, rankings did not. Models trained on CHB-MIT reached on Siena the macro F1 of models trained on Siena itself, but the order of the operators on CHB-MIT, on Siena and after transfer were unrelated (mean Kendall tau between -0.11 and 0.07). Score fusion is the clearest case: first in 82 percent of Siena's own runs, it was first in 11 percent of the transfer runs and among the weaker operators on CHB-MIT. Its advantage on Siena is therefore a property of training and testing on Siena, not a property of the operator. Part of it is inherited from its spectrogram branch: on Siena the spectrogram network alone (0.660 macro F1) already had a higher mean than every jointly trained operator, and score fusion averages it with the waveform network, with a weight chosen on Siena's validation subjects. Whether end-to-end training of the joint operators overfits Siena's 14 subjects, which training the branches independently would avoid, we did not test. The two datasets differ in age, seizure types, recording protocol and mains frequency (Section 2.1), so we do not attribute the difference to any one of these.

4.7 Statistical significance, practical size and ranking stability

The three properties we kept apart gave different answers, and reporting any one alone would have misled. On Siena score fusion was first in four of five runs, a stable ranking, yet its advantage was not significant and could not be shown to exceed or fall below 0.05 macro F1. On CHB-MIT the interaction was significant before correction and with three seeds, yet its practical size was 0.004 macro F1. And on CHB-MIT several pairs of operators were practically equivalent while their order changed from seed to seed, which is what equivalence implies: when two operators are equally good, their order is decided by noise. A stable ranking is not evidence of a meaningful difference, a significant effect is not evidence of a large one, and the absence of a significant difference is not evidence of equivalence.

4.8 What this means for comparisons of fusion operators

We did not find a fusion operator that is better than the others across preprocessing, seeds and datasets, and our results do not support claims that one is. That is itself the main finding: a claim of superiority based on one dataset, one pipeline and one training run, the common design in this literature, would have reached a different conclusion in different runs of our study. Comparisons of this kind should report several seeds and, where feasible, several preprocessing choices; test differences with corrected and equivalence tests rather than naive paired tests; report the rank distribution rather than one ranking; include a parameter-matched unimodal control and a simple classical baseline; and state the margin below which a difference is considered irrelevant before looking at the results.

4.9 What do the results mean clinically?

(Draft. All nine pipelines on CHB-MIT, five seed sets each.)

None of the systems studied here is close to clinical use, and the study was not designed to produce one: the threshold was fixed at 0.5, there was no post-processing, and the models were trained on other people's seizures only. At that operating point the operators detected 68 to 79 percent of CHB-MIT's seizures with 5.2 to 7.7 false alarms per hour, that is 125 to 185 false alarms per day of monitoring, and about 2 percent of all detections were seizures. Generic scalp EEG detectors evaluated for clinical use have reported 0.1 to 5 false alarms per hour, most about 0.2 to 0.3, and patient-specific detectors fewer, and a low false alarm rate is regarded as essential for clinical acceptance, and validation standards require it to be reported (Baumgartner et al. 2018; Beniczky and Ryvlin 2018); no person in CHB-MIT reached a mean below one false alarm per hour. The detection threshold trades sensitivity against false alarms, and training or adjusting the detector per patient lowers the false alarm rate (Baumgartner et al. 2018): patient-specific classifiers detected 96 percent of the CHB-MIT seizures with a median of two false detections per day (Shoeb and Guttag 2010). We left per-patient thresholds and post-processing of consecutive windows out on purpose, so that the operators are compared on their raw output, and the absolute numbers should be read in that light. The criteria were fixed before the runs; choosing a threshold or a post-processing rule after seeing the results would select the criterion on the outcome.

Three results nevertheless bear on clinical use. First, the operators that differed by a few hundredths of macro F1 at the window level differed by little more at the seizure level, except early fusion, which detected the most seizures (0.79) at the most false alarms (7.7 per hour); which of these is preferable is a clinical choice between missed seizures and alarm fatigue (Sendelbach and Funk 2013), not a property that window-level metrics reveal. Second, the person mattered more than any modelling choice: mean sensitivity ranged from 0.24 (chb06) to 1.00 across persons and the false alarm rate from 1.3 to 12.3 per hour, while the operators' means spanned 0.68 to 0.79 and 5.2 to 7.7. For a patient, whether the detector works depends far more on who they are than on the fusion operator, which supports patient-specific adaptation over the search for a better generic operator (Shoeb and Guttag 2010; Baumgartner et al. 2018). Third, an artefact removal step that does not reproduce across machines (Section 4.5) is a clinical risk as much as a scientific one: the same model could detect a different set of a patient's seizures depending on where the recording is cleaned.

Siena cannot answer the clinical question of false alarms: its recordings are cut around seizures, so we report only its sensitivity (84 to 87 percent), which is consistent with CHB-MIT's and says nothing about specificity in continuous monitoring. Neither dataset contains the long, artefact-rich ambulatory or intensive-care recordings in which seizure detectors are used, and the persons are few (23 and 14).

4.10 Limitations

(draft; to be completed with the results)

- **Sample size.** Twenty-three CHB-MIT and fourteen Siena subjects make the corrected confidence intervals wide: differences between operators that are small can be neither shown nor excluded at the equivalence margins, so many pairs are neither different nor demonstrably equivalent.
- **One design family.** All operators share one pair of representations (raw waveform and log-magnitude STFT), one backbone family and one parameter budget; other representations (wavelets, other time-frequency transforms) or larger models may behave differently. Widening a single branch to the fusion budget did not change its performance (Section 3.1), but deeper backbones were not tested.
- **Windows, not sessions.** Window-level metrics use non-ictal windows subsampled to four per ictal window, so they do not reflect the true prevalence of seizures; the seizure-level metrics on the full recordings address this but are secondary. A window is labelled ictal if it overlaps a seizure at all; excluding partly ictal windows is implemented but was not run.
- **Siena.** Its recordings are cut around seizures, so false alarms per hour cannot be interpreted and are not reported; the summed recording time (141.0 h by the EDF headers) differs from the dataset description (128.4 h) for reasons that remain unexplained for three patients. Six problems in its seizure lists were resolved by decisions documented in Supplementary Table S4.
- **CHB-MIT.** Thirteen of its 198 seizures lie in recordings without the standard montage and are not used; chb24 is not listed in the subject information file and is treated as a distinct person.
- **Artefact removal.** The settings (16 components, frontopolar channels as EOG proxies, no muscle rejection) and the projection of GEDAI's reference onto a bipolar montage are our choices for montages without EOG; GEDAI was designed for referential, high-density recordings. That Infomax's and GEDAI's results do not reproduce across machines is a finding of this study (Sections 2.7 and 4.5); for its own results, window-level metrics rest on one cleaning, but the seizure-level metrics of P6a differ from the training-time cleaning in 8 of 670 recordings.
- **Device.** All multiverse runs used one GPU; GPU and CPU results at the same seeds differ by an amount comparable to the difference between seed sets, so absolute numbers are not interchangeable with the CPU results of earlier drafts.
- **Offline processing.** Filters and artefact cleaning see the complete recording (Section 2.2), so the results describe offline preprocessing, not a causal, real-time detector.
- **No tuning.** One fixed training configuration was used for every model; an operator might gain more than another from tuning, which this design does not test.
- **Two datasets.** CHB-MIT and Siena differ in population, acquisition and recording protocol at once, so the transfer experiment measures their combined shift and cannot attribute a change in performance or ranking to any one of these differences.
- **Seeds.** Five seed sets bound the seed variance only roughly; the rank analyses report it rather than average it away.
- **Literature.** Title and abstract screening of the systematic search was done by an AI assistant (agreement with an independent AI pass, kappa 0.81); a human screening and the full texts of the remaining candidates are in progress, and claims marked [abstract] in Section 1 rest on abstracts.

Use of AI assistance

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Supplementary Material

Table S1: Artefact removal in P6: signal retention and compute. Per recording of CHB-MIT with selected windows (670), on the 0.5 to 40 Hz signal. Power kept = signal power after removal / before. Extended Infomax and AMICA share 16 components, fitting on every 8th sample and the same automatic ocular component rule; GEDAI selects what to remove by its own criterion. Compute on a laptop CPU (Intel Core Ultra 7 155U), six recordings in parallel.

method	recordings	power kept, median	IQR	min	share < 0.5	components removed, mean	seconds per recording, median	CPU hours, total
P6a Extended Infomax	670	0.784	0.531-0.899	0.090	0.222	1.491	33	8.5
P6b GEDAI, conservative (used)	670	0.999	0.717-0.999	0.041	0.206		18	4.7
P6c AMICA	670	0.784	0.535-0.885	0.076	0.228	1.531	213	73.1
GEDAI, default (not used)	670	0.658	0.317-0.999	0.041	0.433		22	5.6

Table S2: GEDAI preset, chosen on signal preservation only. Six CHB-MIT subjects (chb01, chb03, chb05, chb08, chb12, chb15), the same windows under both presets. Power kept per recording and per window (relative to the same 0.5 to 40 Hz signal); no classification result was used for the choice (`src/gedai_qc.py`).

preset	unit	n	p5	p25	median	p75	share < 0.5	share < 0.1
default (auto)	recording	199	0.197	0.324	0.843	0.999	0.387	0.000
default (auto)	ictal window	591	0.241	0.367	0.779	1.000	0.333	0.005
default (auto)	non-ictal window	2364	0.263	0.395	0.999	0.999	0.328	0.003
conservative (auto-), used	recording	199	0.263	0.957	0.999	0.999	0.166	0.000
conservative (auto-), used	ictal window	591	0.313	0.999	1.000	1.000	0.134	0.000
conservative (auto-), used	non-ictal window	2364	0.331	0.999	0.999	1.000	0.126	0.002

Table S3: P4: technical artefact rejection per person. A window is rejected if a channel is flat (standard deviation below 1 μV) or holds one constant value for at least 0.5 s, on the unfiltered signal. Rejected windows are removed from training and validation only; test windows are unchanged. Rejection rates are given for ictal and non-ictal windows separately. chb01 includes chb21, the same person.

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb01	370	74	0	296	1	0.0	0.3
chb02	100	20	2	80	2	10.0	2.5
chb03	225	45	7	180	1	15.6	0.6
chb04	210	42	5	168	5	11.9	3.0
chb05	310	62	0	248	0	0.0	0.0
chb06	120	24	1	96	0	4.2	0.0
chb07	180	36	2	144	1	5.6	0.7
chb08	485	97	4	388	3	4.1	0.8
chb09	155	31	3	124	0	9.7	0.0
chb10	245	49	11	196	4	22.4	2.0
chb11	415	83	1	332	0	1.2	0.0
chb12	615	123	1	492	7	0.8	1.4

person	windows	ictal	ictal rejected	non-ictal	non-ictal rejected	ictal rejected %	non-ictal rejected %
chb13	335	67	0	268	5	0.0	1.9
chb14	120	24	0	96	0	0.0	0.0
chb15	1070	214	0	856	1	0.0	0.1
chb16	70	14	0	56	1	0.0	1.8
chb17	160	32	0	128	6	0.0	4.7
chb18	190	38	1	152	0	2.6	0.0
chb19	135	27	3	108	0	11.1	0.0
chb20	180	36	1	144	2	2.8	1.4
chb22	120	24	0	96	0	0.0	0.0
chb23	245	49	0	196	0	0.0	0.0
chb24	325	65	0	260	0	0.0	0.0
all	6380	1276	42	5104	39	3.3	0.8

Table S4: Siena: metadata problems and the decisions taken. Every departure from the published seizure lists, with the decision used in the main analysis, its kind (a correction verified against the recording, or a decision rule applied without such a check), how it was checked, and how much the alternative would change the windows. Other, purely textual fixes (a space inside a time in PN10's list, misspelt file names, PN12's missing registration times taken from the same file's first block, recordings past midnight read modulo 24 h) are listed in `docs/SIENA.md`, Section 4. Recording time is taken from the EDF headers, 141.0 h; the dataset's description gives 128.4 h, a difference explained for PN14 (the start time above) but not for PN03, PN10 and PN12, whose files contain no padding (`docs/SIENA.md`, Section 8).

item	recording / seizure	problem in the metadata	used	type
1	PN00-3, seizure 3	listed end 19.29.29 lies 1916 s after the recording ends (18.57.33)	end read as 18.29.29 (a 60 s seizure)	verified correction
2	PN05-3, seizure 3	registration start: list 06.01.23, EDF header 06.01.13	EDF header start time, as for every recording	decision rule (not a verified correction)
3	PN10-3, seizure 3	two onsets: 15.43.53 (clinical), 15.43.59 (electrical)	electrical onset (advisor)	decision
4	PN10-4.5.6, seizure 6	only a clinical onset (15.18.26)	used as listed, flagged in the corpus's seizure notes	decision
5	PN10-2, seizure 2	end "11.41.04 opure 11.40.43" (Italian "or")	first value, 11.41.04	decision rule
6	PN14-3	registration start: list 16.17.45, EDF header 19.17.45	EDF header start time	verified correction

How each decision was checked, and its effect:

item	how it was checked	remaining uncertainty	effect on the windows
1	signal: the right frontotemporal 3-20 Hz ictal rhythm (FP2-F8, F8-T8) falls below a quarter of its peak 14 s after 18.29.29; the patient's other four seizures end 0 to 13 s after their listed ends by the same rule	offset from an automatic rule on one channel pair	alternative (exclude the seizure and the rest of the recording): 30 instead of 37 ictal windows for PN00

item	how it was checked	remaining uncertainty	effect on the windows
2	none possible: the header defines the first sample and 39 of 41 recordings agree with their list; the list's value fits the 20 s end offset of 18 other recordings, the header's does not	10 s in the seizure's position	4 ictal windows either way, shifted by one: 2 window labels differ
3	CHB-MIT annotates electrographic onsets; the electrical onset keeps the two corpora consistent	none (both listed)	clinical onset instead: 48 instead of 47 ictal windows for PN10 (1 label)
4	-	electrical onset unknown; in seizure 3 it followed the clinical one by 6 s	at most about one window, by analogy with seizure 3
5	-	21 s in the seizure's end	second value instead: 45 instead of 47 ictal windows for PN10
6	header start + duration (41,995 s) gives the listed end time 06.57.40 exactly; the list's start would make the recording three hours longer than the file	none	seizure at 1 h 52 min into the recording, not 4 h 52 min

Table S5: Artefact removal: power kept per window, by band and class. What each method removes from the 0.5 to 40 Hz signal, per 10 s window (cleaned power divided by the power of the same window before cleaning, summed over channels), for ictal and non-ictal windows, overall and in five bands (delta 0.5-4, theta 4-8, alpha 8-13, beta 13-30, low gamma 30-40 Hz; medians). The second part compares the classes within recordings: per recording the median of each class, their difference, and a Wilcoxon signed-rank test over the recordings that hold both. A difference between the classes over the whole corpus that is absent within recordings reflects which recordings a method cleans heavily, not a preference for one class (`src/p6_signal_effects.py`). GEDAI's default preset was applied to CHB-MIT only.

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
chbmit	Extended Infomax	ictal	1276	0.872	0.092	0.855	0.905	0.929	0.933	0.935
chbmit	Extended Infomax	non-ictal	5104	0.854	0.187	0.835	0.893	0.911	0.927	0.933
chbmit	GEDAI conservative (used)	ictal	1276	1.000	0.134	0.999	1.000	1.000	1.000	1.000
chbmit	GEDAI conservative (used)	non-ictal	5104	0.999	0.167	0.999	1.000	1.000	1.000	1.000
chbmit	GEDAI default (not used)	ictal	1276	0.625	0.448	0.583	0.739	0.836	0.882	0.909
chbmit	GEDAI default (not used)	non-ictal	5104	0.925	0.372	0.920	0.966	0.983	0.991	0.994
chbmit	AMICA	ictal	1276	0.873	0.093	0.851	0.909	0.922	0.929	0.933
chbmit	AMICA	non-ictal	5104	0.839	0.191	0.821	0.882	0.901	0.921	0.930
siena	Extended Infomax	ictal	328	0.682	0.271	0.609	0.759	0.847	0.900	0.941
siena	Extended Infomax	non-ictal	1312	0.657	0.370	0.520	0.714	0.862	0.906	0.914
siena	GEDAI conservative (used)	ictal	328	0.679	0.427	0.694	0.745	0.850	0.833	0.944

dataset	method	windows	n	power kept, median	share < 0.5	delta	theta	alpha	beta	low gamma
siena	GEDAI conservative (used)	non-ictal	1312	0.997	0.383	0.997	1.000	1.000	1.000	1.000
siena	AMICA	ictal	328	0.723	0.232	0.660	0.823	0.887	0.929	0.959
siena	AMICA	non-ictal	1312	0.719	0.334	0.627	0.800	0.915	0.936	0.939

Within recordings:

dataset	method	recordings	ictal - non-ictal, median	share with ictal lower	Wilcoxon p
chbmit	Extended Infomax	137	0.0098	0.2847	0.0001
chbmit	GEDAI conservative (used)	137	0.0001	0.3212	0.0016
chbmit	GEDAI default (not used)	137	-0.0000	0.5036	0.0363
chbmit	AMICA	137	0.0170	0.2847	0.0000
siena	Extended Infomax	41	-0.0050	0.5122	0.6769
siena	GEDAI conservative (used)	41	-0.0000	0.5610	0.6815
siena	AMICA	41	0.0000	0.4390	0.8790

Table S6: Window-level performance per pipeline and operator. Mean and SD over the five seed sets of each run's fold mean, for every pipeline and fusion operator (`python -m src.make_results`, `results_v2/results/main_*.csv`). **CHB-MIT, macro F1** (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.678 ±0.013	0.679 ±0.012	0.671 ±0.013	**0.683** ±0.020	0.661 ±0.014
P1	0.684 ±0.008	0.665 ±0.015	**0.691** ±0.005	0.687 ±0.008	0.672 ±0.017
P2	0.680 ±0.014	**0.688** ±0.009	0.680 ±0.021	0.687 ±0.009	0.675 ±0.008
P3	0.679 ±0.011	0.677 ±0.017	0.676 ±0.015	**0.689** ±0.015	0.678 ±0.014
P4	0.688 ±0.010	0.678 ±0.008	**0.694** ±0.012	0.670 ±0.010	0.660 ±0.024
P5	0.678 ±0.012	0.684 ±0.019	0.690 ±0.021	**0.693** ±0.005	0.681 ±0.008
P6a	0.672 ±0.014	0.679 ±0.014	**0.692** ±0.018	0.677 ±0.013	0.675 ±0.012
P6b	0.662 ±0.012	0.686 ±0.018	**0.691** ±0.012	0.681 ±0.020	0.682 ±0.028
P6c	0.671 ±0.007	0.688 ±0.019	**0.688** ±0.013	0.679 ±0.014	0.674 ±0.010

CHB-MIT, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.632 ±0.016	0.656 ±0.018	**0.662** ±0.019	0.655 ±0.015	0.654 ±0.011
P1	0.623 ±0.015	0.665 ±0.011	**0.673** ±0.009	0.665 ±0.017	0.667 ±0.019
P2	0.618 ±0.021	0.668 ±0.007	**0.683** ±0.016	0.669 ±0.013	0.657 ±0.021
P3	0.607 ±0.013	0.657 ±0.012	0.655 ±0.016	**0.658** ±0.031	0.651 ±0.011
P4	0.628 ±0.018	0.672 ±0.018	**0.673** ±0.013	0.650 ±0.026	0.651 ±0.018
P5	0.608 ±0.019	0.642 ±0.017	**0.664** ±0.022	0.660 ±0.012	0.662 ±0.005
P6a	0.619 ±0.021	0.653 ±0.028	**0.668** ±0.015	0.652 ±0.017	0.657 ±0.010
P6b	0.564 ±0.010	0.635 ±0.011	**0.643** ±0.018	0.636 ±0.015	0.631 ±0.015
P6c	0.617 ±0.009	0.658 ±0.023	**0.671** ±0.010	0.654 ±0.018	0.668 ±0.012

CHB-MIT, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	**0.684** ±0.009	0.681 ±0.010	0.674 ±0.010	0.684 ±0.016	0.667 ±0.014
P1	0.684 ±0.006	0.669 ±0.010	**0.685** ±0.008	0.684 ±0.008	0.680 ±0.015
P2	0.684 ±0.010	0.682 ±0.006	0.679 ±0.022	**0.685** ±0.009	0.678 ±0.010
P3	**0.687** ±0.010	0.676 ±0.015	0.676 ±0.012	0.687 ±0.013	0.683 ±0.011
P4	**0.690** ±0.008	0.677 ±0.006	0.687 ±0.008	0.667 ±0.011	0.663 ±0.020
P5	0.686 ±0.015	0.683 ±0.014	0.684 ±0.019	**0.694** ±0.008	0.687 ±0.008
P6a	0.676 ±0.014	0.675 ±0.014	**0.688** ±0.017	0.680 ±0.006	0.679 ±0.014
P6b	0.670 ±0.009	0.682 ±0.017	**0.688** ±0.012	0.680 ±0.016	0.686 ±0.017
P6c	0.677 ±0.008	0.682 ±0.017	**0.686** ±0.009	0.680 ±0.015	0.681 ±0.010

Siena, macro F1 (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.601 ±0.021	0.616 ±0.023	0.608 ±0.016	0.598 ±0.020	**0.639** ±0.008
P1	0.616 ±0.024	0.637 ±0.012	0.623 ±0.020	0.628 ±0.026	**0.665** ±0.019
P2	0.610 ±0.016	0.639 ±0.014	0.636 ±0.018	0.625 ±0.016	**0.674** ±0.012
P3	0.613 ±0.014	0.652 ±0.039	0.620 ±0.013	0.613 ±0.010	**0.664** ±0.029
P4	0.614 ±0.020	0.637 ±0.012	0.621 ±0.019	0.627 ±0.025	**0.666** ±0.020
P5	0.637 ±0.029	0.642 ±0.017	0.650 ±0.015	0.652 ±0.008	**0.690** ±0.031
P6a	0.619 ±0.018	0.622 ±0.019	0.621 ±0.048	0.611 ±0.013	**0.672** ±0.018
P6b	0.614 ±0.010	0.681 ±0.012	**0.684** ±0.015	0.652 ±0.018	0.676 ±0.027
P6c	0.629 ±0.027	0.638 ±0.010	0.613 ±0.024	0.621 ±0.013	**0.683** ±0.026

Siena, AUPRC (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.453 ±0.018	0.556 ±0.023	0.527 ±0.018	0.522 ±0.003	**0.571** ±0.014
P1	0.498 ±0.025	0.613 ±0.039	0.615 ±0.019	0.601 ±0.021	**0.649** ±0.024
P2	0.494 ±0.036	0.624 ±0.035	0.594 ±0.019	0.599 ±0.020	**0.642** ±0.007
P3	0.492 ±0.020	0.633 ±0.043	0.611 ±0.013	0.588 ±0.032	**0.650** ±0.029
P4	0.497 ±0.026	0.613 ±0.039	0.614 ±0.017	0.601 ±0.022	**0.648** ±0.025
P5	0.540 ±0.021	0.611 ±0.027	0.610 ±0.019	0.582 ±0.023	**0.641** ±0.027
P6a	0.510 ±0.030	0.587 ±0.042	0.574 ±0.045	0.575 ±0.041	**0.621** ±0.028
P6b	0.460 ±0.020	0.590 ±0.017	0.585 ±0.022	0.571 ±0.028	**0.637** ±0.026
P6c	0.522 ±0.034	0.563 ±0.028	0.585 ±0.015	0.578 ±0.034	**0.618** ±0.034

Siena, balanced accuracy (5 seed sets; best operator of each pipeline in bold)

pipeline	early	late	gated	attention	score
P0	0.621 ±0.021	0.663 ±0.018	0.652 ±0.011	0.639 ±0.013	**0.665** ±0.007
P1	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.024	**0.689** ±0.022
P2	0.645 ±0.024	0.680 ±0.008	0.684 ±0.012	0.674 ±0.014	**0.696** ±0.010
P3	0.641 ±0.018	**0.693** ±0.031	0.671 ±0.010	0.660 ±0.009	0.684 ±0.021
P4	0.643 ±0.022	0.676 ±0.014	0.673 ±0.018	0.673 ±0.025	**0.691** ±0.024
P5	0.661 ±0.023	0.688 ±0.015	0.691 ±0.011	0.687 ±0.010	**0.710** ±0.038
P6a	0.645 ±0.016	0.668 ±0.016	0.664 ±0.038	0.658 ±0.015	**0.682** ±0.011
P6b	0.622 ±0.011	0.688 ±0.017	**0.694** ±0.018	0.676 ±0.015	0.691 ±0.022
P6c	0.654 ±0.019	0.676 ±0.006	0.660 ±0.019	0.663 ±0.020	**0.697** ±0.018

Table S7: All controls. The six controls over the same complete runs as Table 4 (mean and SD over the 45 runs of each dataset, fold means within a run; Brier: lower is better). Parameters with 18 input channels; the classical models are logistic regressions on 18 log variances (logvar) and on seven features per channel (shallow), with an intercept (`results_v2/results/controls_*.csv`).

control	parameters	CHB-MIT macro F1	CHB-MIT AUPRC	CHB-MIT bal. acc.	CHB-MIT Brier	Siena macro F1	Siena AUPRC	Siena bal. acc.	Siena Brier
raw1d	28,850	0.663 ±0.014	0.604 ±0.019	0.678 ±0.013	0.332 ±0.018	0.609 ±0.034	0.565 ±0.040	0.659 ±0.024	0.443 ±0.061
spec2d	34,546	0.679 ±0.010	0.598 ±0.017	0.676 ±0.010	0.309 ±0.011	0.660 ±0.022	0.576 ±0.031	0.680 ±0.019	0.350 ±0.030
raw1d_wide	103,712	0.662 ±0.012	0.605 ±0.017	0.673 ±0.010	0.328 ±0.016	0.612 ±0.028	0.577 ±0.033	0.657 ±0.019	0.440 ±0.056
spec2d_wide	102,236	0.683 ±0.014	0.618 ±0.016	0.680 ±0.013	0.302 ±0.014	0.663 ±0.026	0.581 ±0.027	0.681 ±0.021	0.352 ±0.038
logvar	19	0.631 ±0.009	0.577 ±0.016	0.708 ±0.006	0.402 ±0.009	0.630 ±0.034	0.572 ±0.036	0.705 ±0.027	0.455 ±0.036
shallow	127	0.690 ±0.008	0.640 ±0.008	0.739 ±0.007	0.339 ±0.014	0.720 ±0.019	0.667 ±0.015	0.770 ±0.017	0.338 ±0.020

Table S8: Paired differences between operators, with confidence intervals and equivalence. For every operator pair in every pipeline: the mean paired difference in macro F1, its 90 percent confidence interval with the corrected variance (the interval used by the two one-sided tests), the corrected and the Holm-adjusted p of a difference, the smallest equivalence margin the data can establish ($\delta_{\min}$), and the outcome at the margins of ±0.05 and ±0.02 (`results_v2/multiverse/*_cuda/equivalence*.csv`; Figure 3). **CHB-MIT** (difference: first minus second operator, macro F1; fold values averaged over the seed sets)

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P0	early-late	-0.001	[-0.034, +0.033]	0.973	1.000	0.034	yes	no
P0	early-gated	+0.007	[-0.023, +0.037]	0.678	1.000	0.037	yes	no
P0	early-attention	-0.004	[-0.039, +0.030]	0.827	1.000	0.039	yes	no
P0	early-score	+0.018	[-0.014, +0.050]	0.353	1.000	0.050	yes	no
P0	late-gated	+0.008	[-0.012, +0.028]	0.491	1.000	0.028	yes	no
P0	late-attention	-0.004	[-0.020, +0.012]	0.692	1.000	0.020	yes	yes
P0	late-score	+0.018	[-0.009, +0.046]	0.266	1.000	0.046	yes	no
P0	gated-attention	-0.012	[-0.038, +0.015]	0.458	1.000	0.038	yes	no
P0	gated-score	+0.010	[-0.020, +0.041]	0.561	1.000	0.041	yes	no
P0	attention-score	+0.022	[-0.006, +0.050]	0.194	1.000	0.050	no	no
P1	early-late	+0.018	[-0.022, +0.059]	0.442	1.000	0.059	no	no
P1	early-gated	-0.008	[-0.041, +0.026]	0.701	1.000	0.041	yes	no
P1	early-attention	-0.004	[-0.041, +0.034]	0.866	1.000	0.041	yes	no
P1	early-score	+0.011	[-0.023, +0.046]	0.577	1.000	0.046	yes	no
P1	late-gated	-0.026	[-0.052, -0.000]	0.097	0.896	0.052	no	no
P1	late-attention	-0.022	[-0.043, -0.001]	0.090	0.896	0.043	yes	no
P1	late-score	-0.007	[-0.037, +0.023]	0.692	1.000	0.037	yes	no
P1	gated-attention	+0.004	[-0.017, +0.025]	0.752	1.000	0.025	yes	no
P1	gated-score	+0.019	[-0.013, +0.051]	0.326	1.000	0.051	no	no
P1	attention-score	+0.015	[-0.015, +0.045]	0.398	1.000	0.045	yes	no
P2	early-late	-0.009	[-0.046, +0.029]	0.689	1.000	0.046	yes	no
P2	early-gated	-0.000	[-0.034, +0.033]	0.982	1.000	0.034	yes	no
P2	early-attention	-0.007	[-0.041, +0.026]	0.705	1.000	0.041	yes	no
P2	early-score	+0.004	[-0.027, +0.035]	0.813	1.000	0.035	yes	no
P2	late-gated	+0.008	[-0.017, +0.034]	0.572	1.000	0.034	yes	no
P2	late-attention	+0.001	[-0.020, +0.023]	0.911	1.000	0.023	yes	no
P2	late-score	+0.013	[-0.018, +0.045]	0.478	1.000	0.045	yes	no
P2	gated-attention	-0.007	[-0.038, +0.024]	0.702	1.000	0.038	yes	no
P2	gated-score	+0.005	[-0.024, +0.034]	0.779	1.000	0.034	yes	no
P2	attention-score	+0.012	[-0.020, +0.044]	0.532	1.000	0.044	yes	no
P3	early-late	+0.002	[-0.035, +0.039]	0.919	1.000	0.039	yes	no
P3	early-gated	+0.003	[-0.036, +0.042]	0.889	1.000	0.042	yes	no
P3	early-attention	-0.010	[-0.050, +0.031]	0.684	1.000	0.050	yes	no
P3	early-score	+0.001	[-0.032, +0.035]	0.954	1.000	0.035	yes	no
P3	late-gated	+0.001	[-0.021, +0.023]	0.938	1.000	0.023	yes	no
P3	late-attention	-0.012	[-0.032, +0.008]	0.326	1.000	0.032	yes	no
P3	late-score	-0.001	[-0.029, +0.027]	0.948	1.000	0.029	yes	no
P3	gated-attention	-0.013	[-0.038, +0.012]	0.387	1.000	0.038	yes	no
P3	gated-score	-0.002	[-0.033, +0.028]	0.907	1.000	0.033	yes	no
P3	attention-score	+0.011	[-0.019, +0.041]	0.546	1.000	0.041	yes	no
P4	early-late	+0.010	[-0.018, +0.038]	0.554	1.000	0.038	yes	no
P4	early-gated	-0.006	[-0.042, +0.030]	0.781	1.000	0.042	yes	no
P4	early-attention	+0.018	[-0.010, +0.046]	0.277	1.000	0.046	yes	no
P4	early-score	+0.029	[-0.007, +0.065]	0.182	1.000	0.065	no	no
P4	late-gated	-0.016	[-0.035, +0.003]	0.160	1.000	0.035	yes	no
P4	late-attention	+0.008	[-0.005, +0.022]	0.305	1.000	0.022	yes	no
P4	late-score	+0.019	[-0.013, +0.051]	0.316	1.000	0.051	no	no
P4	gated-attention	+0.024	[+0.008, +0.041]	0.019	0.192	0.041	yes	no
P4	gated-score	+0.035	[+0.006, +0.064]	0.052	0.469	0.064	no	no
P4	attention-score	+0.011	[-0.013, +0.035]	0.456	1.000	0.035	yes	no
P5	early-late	-0.006	[-0.048, +0.037]	0.817	1.000	0.048	yes	no
P5	early-gated	-0.012	[-0.056, +0.032]	0.640	1.000	0.056	no	no
P5	early-attention	-0.015	[-0.062, +0.032]	0.595	1.000	0.062	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P5	early-score	-0.003	[-0.045, +0.038]	0.897	1.000	0.045	yes	no
P5	late-gated	-0.006	[-0.028, +0.016]	0.623	1.000	0.028	yes	no
P5	late-attention	-0.009	[-0.027, +0.009]	0.398	1.000	0.027	yes	no
P5	late-score	+0.003	[-0.029, +0.035]	0.890	1.000	0.035	yes	no
P5	gated-attention	-0.003	[-0.027, +0.021]	0.849	1.000	0.027	yes	no
P5	gated-score	+0.009	[-0.025, +0.043]	0.655	1.000	0.043	yes	no
P5	attention-score	+0.012	[-0.018, +0.042]	0.511	1.000	0.042	yes	no
P6a	early-late	-0.008	[-0.041, +0.025]	0.680	1.000	0.041	yes	no
P6a	early-gated	-0.021	[-0.052, +0.010]	0.254	1.000	0.052	no	no
P6a	early-attention	-0.006	[-0.040, +0.029]	0.775	1.000	0.040	yes	no
P6a	early-score	-0.003	[-0.032, +0.025]	0.856	1.000	0.032	yes	no
P6a	late-gated	-0.013	[-0.025, -0.001]	0.077	0.768	0.025	yes	no
P6a	late-attention	+0.002	[-0.017, +0.021]	0.851	1.000	0.021	yes	no
P6a	late-score	+0.005	[-0.025, +0.035]	0.782	1.000	0.035	yes	no
P6a	gated-attention	+0.015	[-0.004, +0.034]	0.191	1.000	0.034	yes	no
P6a	gated-score	+0.018	[-0.005, +0.041]	0.194	1.000	0.041	yes	no
P6a	attention-score	+0.003	[-0.027, +0.032]	0.875	1.000	0.032	yes	no
P6b	early-late	-0.025	[-0.064, +0.015]	0.291	1.000	0.064	no	no
P6b	early-gated	-0.029	[-0.066, +0.008]	0.187	1.000	0.066	no	no
P6b	early-attention	-0.019	[-0.059, +0.020]	0.403	1.000	0.059	no	no
P6b	early-score	-0.020	[-0.054, +0.014]	0.314	1.000	0.054	no	no
P6b	late-gated	-0.004	[-0.026, +0.017]	0.726	1.000	0.026	yes	no
P6b	late-attention	+0.005	[-0.021, +0.032]	0.730	1.000	0.032	yes	no
P6b	late-score	+0.005	[-0.030, +0.039]	0.826	1.000	0.039	yes	no
P6b	gated-attention	+0.010	[-0.017, +0.036]	0.531	1.000	0.036	yes	no
P6b	gated-score	+0.009	[-0.028, +0.046]	0.683	1.000	0.046	yes	no
P6b	attention-score	-0.001	[-0.030, +0.028]	0.960	1.000	0.030	yes	no
P6c	early-late	-0.017	[-0.063, +0.029]	0.532	1.000	0.063	no	no
P6c	early-gated	-0.017	[-0.051, +0.016]	0.378	1.000	0.051	no	no
P6c	early-attention	-0.008	[-0.042, +0.025]	0.676	1.000	0.042	yes	no
P6c	early-score	-0.003	[-0.035, +0.029]	0.872	1.000	0.035	yes	no
P6c	late-gated	-0.000	[-0.021, +0.020]	0.976	1.000	0.021	yes	no
P6c	late-attention	+0.009	[-0.012, +0.029]	0.470	1.000	0.029	yes	no
P6c	late-score	+0.014	[-0.026, +0.054]	0.555	1.000	0.054	no	no
P6c	gated-attention	+0.009	[-0.008, +0.026]	0.374	1.000	0.026	yes	no
P6c	gated-score	+0.014	[-0.019, +0.048]	0.469	1.000	0.048	yes	no
P6c	attention-score	+0.005	[-0.026, +0.037]	0.776	1.000	0.037	yes	no

Siena (difference: first minus second operator, macro F1; fold values averaged over the seed sets)

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P0	early-late	-0.015	[-0.056, +0.025]	0.513	1.000	0.056	no	no
P0	early-gated	-0.007	[-0.062, +0.048]	0.820	1.000	0.062	no	no
P0	early-attention	+0.003	[-0.048, +0.055]	0.910	1.000	0.055	no	no
P0	early-score	-0.038	[-0.081, +0.004]	0.132	1.000	0.081	no	no
P0	late-gated	+0.008	[-0.025, +0.041]	0.666	1.000	0.041	yes	no
P0	late-attention	+0.019	[-0.016, +0.053]	0.355	1.000	0.053	no	no
P0	late-score	-0.023	[-0.058, +0.012]	0.269	1.000	0.058	no	no
P0	gated-attention	+0.011	[-0.007, +0.028]	0.316	1.000	0.028	yes	no
P0	gated-score	-0.031	[-0.054, -0.009]	0.029	0.257	0.054	no	no
P0	attention-score	-0.042	[-0.066, -0.017]	0.009	0.095	0.066	no	no
P1	early-late	-0.020	[-0.095, +0.055]	0.643	1.000	0.095	no	no
P1	early-gated	-0.007	[-0.104, +0.091]	0.905	1.000	0.104	no	no
P1	early-attention	-0.012	[-0.106, +0.083]	0.833	1.000	0.106	no	no
P1	early-score	-0.049	[-0.106, +0.008]	0.149	1.000	0.106	no	no
P1	late-gated	+0.013	[-0.029, +0.056]	0.580	1.000	0.056	no	no
P1	late-attention	+0.009	[-0.028, +0.045]	0.685	1.000	0.045	yes	no
P1	late-score	-0.029	[-0.083, +0.026]	0.367	1.000	0.083	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P1	gated-attention	-0.005	[-0.039, +0.030]	0.806	1.000	0.039	yes	no
P1	gated-score	-0.042	[-0.110, +0.026]	0.291	1.000	0.110	no	no
P1	attention-score	-0.037	[-0.098, +0.024]	0.297	1.000	0.098	no	no
P2	early-late	-0.029	[-0.100, +0.042]	0.485	1.000	0.100	no	no
P2	early-gated	-0.026	[-0.107, +0.055]	0.581	1.000	0.107	no	no
P2	early-attention	-0.015	[-0.100, +0.071]	0.766	1.000	0.100	no	no
P2	early-score	-0.064	[-0.112, -0.016]	0.034	0.339	0.112	no	no
P2	late-gated	+0.003	[-0.023, +0.029]	0.844	1.000	0.029	yes	no
P2	late-attention	+0.014	[-0.013, +0.042]	0.379	1.000	0.042	yes	no
P2	late-score	-0.035	[-0.097, +0.026]	0.327	1.000	0.097	no	no
P2	gated-attention	+0.011	[-0.017, +0.040]	0.494	1.000	0.040	yes	no
P2	gated-score	-0.038	[-0.106, +0.029]	0.335	1.000	0.106	no	no
P2	attention-score	-0.050	[-0.127, +0.027]	0.274	1.000	0.127	no	no
P3	early-late	-0.038	[-0.110, +0.034]	0.362	1.000	0.110	no	no
P3	early-gated	-0.007	[-0.115, +0.102]	0.917	1.000	0.115	no	no
P3	early-attention	+0.000	[-0.093, +0.094]	0.994	1.000	0.094	no	no
P3	early-score	-0.050	[-0.127, +0.026]	0.261	1.000	0.127	no	no
P3	late-gated	+0.032	[-0.024, +0.088]	0.331	1.000	0.088	no	no
P3	late-attention	+0.039	[-0.011, +0.088]	0.189	1.000	0.088	no	no
P3	late-score	-0.012	[-0.042, +0.018]	0.483	1.000	0.042	yes	no
P3	gated-attention	+0.007	[-0.035, +0.048]	0.773	1.000	0.048	yes	no
P3	gated-score	-0.044	[-0.103, +0.015]	0.207	1.000	0.103	no	no
P3	attention-score	-0.051	[-0.100, -0.002]	0.089	0.893	0.100	no	no
P4	early-late	-0.022	[-0.096, +0.052]	0.607	1.000	0.096	no	no
P4	early-gated	-0.007	[-0.101, +0.087]	0.901	1.000	0.101	no	no
P4	early-attention	-0.012	[-0.106, +0.081]	0.817	1.000	0.106	no	no
P4	early-score	-0.051	[-0.108, +0.005]	0.130	1.000	0.108	no	no
P4	late-gated	+0.015	[-0.026, +0.056]	0.521	1.000	0.056	no	no
P4	late-attention	+0.010	[-0.027, +0.046]	0.650	1.000	0.046	yes	no
P4	late-score	-0.029	[-0.082, +0.023]	0.341	1.000	0.082	no	no
P4	gated-attention	-0.006	[-0.040, +0.028]	0.771	1.000	0.040	yes	no
P4	gated-score	-0.045	[-0.111, +0.022]	0.254	1.000	0.111	no	no
P4	attention-score	-0.039	[-0.099, +0.021]	0.269	1.000	0.099	no	no
P5	early-late	-0.005	[-0.071, +0.061]	0.895	1.000	0.071	no	no
P5	early-gated	-0.013	[-0.086, +0.061]	0.767	1.000	0.086	no	no
P5	early-attention	-0.015	[-0.080, +0.051]	0.700	1.000	0.080	no	no
P5	early-score	-0.053	[-0.112, +0.007]	0.142	1.000	0.112	no	no
P5	late-gated	-0.008	[-0.040, +0.025]	0.691	1.000	0.040	yes	no
P5	late-attention	-0.010	[-0.058, +0.038]	0.727	1.000	0.058	no	no
P5	late-score	-0.048	[-0.097, +0.002]	0.111	1.000	0.097	no	no
P5	gated-attention	-0.002	[-0.045, +0.040]	0.931	1.000	0.045	yes	no
P5	gated-score	-0.040	[-0.084, +0.004]	0.132	1.000	0.084	no	no
P5	attention-score	-0.038	[-0.084, +0.008]	0.167	1.000	0.084	no	no
P6a	early-late	-0.003	[-0.074, +0.067]	0.933	1.000	0.074	no	no
P6a	early-gated	-0.002	[-0.049, +0.044]	0.936	1.000	0.049	yes	no
P6a	early-attention	+0.008	[-0.052, +0.067]	0.824	1.000	0.067	no	no
P6a	early-score	-0.053	[-0.113, +0.007]	0.139	1.000	0.113	no	no
P6a	late-gated	+0.001	[-0.044, +0.046]	0.960	1.000	0.046	yes	no
P6a	late-attention	+0.011	[-0.019, +0.041]	0.524	1.000	0.041	yes	no
P6a	late-score	-0.050	[-0.148, +0.048]	0.385	1.000	0.148	no	no
P6a	gated-attention	+0.010	[-0.025, +0.044]	0.625	1.000	0.044	yes	no
P6a	gated-score	-0.051	[-0.132, +0.030]	0.282	1.000	0.132	no	no
P6a	attention-score	-0.061	[-0.144, +0.022]	0.218	1.000	0.144	no	no
P6b	early-late	-0.067	[-0.110, -0.023]	0.018	0.178	0.110	no	no
P6b	early-gated	-0.070	[-0.124, -0.015]	0.042	0.338	0.124	no	no
P6b	early-attention	-0.038	[-0.077, +0.001]	0.106	0.745	0.077	no	no
P6b	early-score	-0.062	[-0.105, -0.019]	0.024	0.219	0.105	no	no
P6b	late-gated	-0.003	[-0.021, +0.015]	0.786	1.000	0.021	yes	no
P6b	late-attention	+0.029	[-0.006, +0.064]	0.170	1.000	0.064	no	no
P6b	late-score	+0.005	[-0.033, +0.043]	0.819	1.000	0.043	yes	no
P6b	gated-attention	+0.031	[-0.015, +0.078]	0.255	1.000	0.078	no	no

pipeline	pair	difference	90% CI	p corrected	p Holm	$\delta_{\min}$	equivalent ±0.05	equivalent ±0.02
P6b	gated-score	+0.008	[-0.040, +0.056]	0.779	1.000	0.056	no	no
P6b	attention-score	-0.024	[-0.062, +0.015]	0.293	1.000	0.062	no	no
P6c	early-late	-0.009	[-0.057, +0.038]	0.738	1.000	0.057	no	no
P6c	early-gated	+0.016	[-0.045, +0.077]	0.656	1.000	0.077	no	no
P6c	early-attention	+0.007	[-0.048, +0.063]	0.817	1.000	0.063	no	no
P6c	early-score	-0.054	[-0.114, +0.006]	0.133	1.000	0.114	no	no
P6c	late-gated	+0.025	[-0.016, +0.066]	0.304	1.000	0.066	no	no
P6c	late-attention	+0.017	[-0.010, +0.044]	0.297	1.000	0.044	yes	no
P6c	late-score	-0.045	[-0.111, +0.022]	0.254	1.000	0.111	no	no
P6c	gated-attention	-0.008	[-0.039, +0.022]	0.636	1.000	0.039	yes	no
P6c	gated-score	-0.070	[-0.146, +0.007]	0.131	1.000	0.146	no	no
P6c	attention-score	-0.061	[-0.129, +0.006]	0.131	1.000	0.129	no	no

Supplementary Note 1. The Bonn corpus (earlier stage)

An earlier stage of this project compared the same operators and controls on the Bonn corpus (Andrzejak et al. 2001): five sets of 100 single-channel segments (23.6 s, 173.61 Hz); A and B scalp recordings of healthy volunteers, C and D interictal and E ictal intracranial recordings. Three label groupings were used: T1, three classes (A,B / C,D / E); T2, intracranial only (C,D vs E); T3, the common binary split (A,B,C,D vs E). Evaluation was 5 x 5 repeated stratified cross-validation over segments: Bonn has no subject identifiers, so a subject-wise split, as in the main study, is not possible. One preprocessing pipeline, CPU, earlier code version; these results are not part of the multiverse.

Model	T1 F1	T1 AUC	T2 F1	T2 AUC	T3 F1	T3 AUC	Params
late	0.974	0.999	0.987	1.000	0.986	1.000	98,571
gated	0.973	0.999	0.987	0.999	0.980	1.000	98,698
attention	0.969	0.997	0.985	0.999	0.977	0.998	98,544
spec2d_wide	0.966	0.998	0.984	0.999	0.982	1.000	97,775
score	0.964	0.998	0.981	0.999	0.976	0.999	59,302
raw1d_wide	0.963	0.997	0.973	0.998	0.976	0.992	99,795
spec2d	0.960	0.998	0.979	0.999	0.972	0.999	32,227
shallow	0.954	0.996	0.964	0.995	0.964	0.995	8
raw1d	0.951	0.996	0.964	0.997	0.969	0.995	27,075
early	0.896	0.977	0.923	0.981	0.899	0.965	97,507
logvar	0.646	0.755	0.933	0.986	0.954	0.992	1

Macro F1 and AUC, mean over the 25 paired measurements (`results_v2/T{1,2,3}_*/perfold.csv`; parameter counts of that code version, single-channel input).

Findings: (1) the naive paired Wilcoxon test found 22, 20 and 13 of 55 pairs significant on T1, T2 and T3; the corrected test found 10, 0 and 0, all ten on T1 involving logvar and none between two deep models (`results_v2/phase0/summary.csv`); (2) late, gated, attention and score fusion were equivalent to each other within about 0.02 to 0.03 macro F1 on T1; (3) early fusion was the one operator that separated, and only downward; (4) T3 is close to saturation: a single-feature logistic regression (logvar) reached 0.954 macro F1 against 0.986 for the best model.